

Certified-Everlasting Quantum NIZK Proofs

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Abstract

We study non-interactive zero-knowledge proofs (NIZKs) for $\mathbf{NP}$ satisfying: 1) statistical soundness, 2) computational zero-knowledge and 3) certified-everlasting zero-knowledge (CE-ZK). The CE-ZK property allows a verifier of a quantum proof to revoke the proof in a way that can be checked (certified) by the prover. Conditioned on successful certification, the verifier's state can be efficiently simulated with only the statement, in a *statistically indistinguishable* way. Our contributions regarding these certified-everlasting NIZKs (CE-NIZKs) are as follows:

- We identify a barrier to obtaining CE-NIZKs in the CRS model via generalizations of known interactive zero-knowledge proofs that satisfy CE-ZK.
- We circumvent this by constructing CE-NIZK from black-box use of NIZK for $\mathbf{NP}$ satisfying certain properties, along with OWFs. As a result, we obtain CE-NIZKs for $\mathbf{NP}$ in the CRS model, based on polynomial hardness of the learning with errors (LWE) assumption.
- In addition, we observe that the aforementioned barrier does not apply to the shared EPR model. We leverage this fact to construct a CE-NIZK for $\mathbf{NP}$ in this model based on any statistical binding hidden-bits generator, which can be based on LWE. The only quantum computation in this protocol involves single-qubit measurements of the shared EPR pairs.

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1 Introduction

1.1 Background

A non-interactive zero-knowledge proof (NIZK) [BFM88] allows a *prover* to demonstrate the validity of a statement to a *verifier* with a *single* message, without revealing any other information. As an example, if Alice knows a solution to a Sudoku puzzle, she can convince Bob that the puzzle is indeed solvable, without leaking the solution itself. Crucially, if the puzzle does not actually have a solution, then a cheating Alice should not be able to fool Bob, a guarantee formally known as *soundness*. On the other hand, the inability of a cheating Bob to learn any additional information is referred to as *zero-knowledge*. This latter property is formalized by the existence of an efficient algorithm called a *simulator*, that can recreate Bob’s view using the statement alone. Intuitively, it captures that whatever Bob learns from a proof, he can compute efficiently without it.

NIZKs have been a cornerstone of cryptography with widespread theoretical and practical impact. For example, they play an important role in constructions of CCA-secure PKE [NY90], digital signatures [BG90], ZAPs [DN00], round-efficient MPC [AJL⁺12], and functional encryption [GGH⁺13] to name a few. On the practical front, they have been instrumental in designing anonymous credentials [BCC⁺09], group signatures [BCC⁺16], cryptocurrencies [SCG⁺14] and verifiable computation [GGP10].

Setup Models. It is well-known that NIZKs suffer a drawback compared to their multi-round (interactive) counterparts. Specifically, they are impossible to obtain in the plain model [GO94]. Fortunately, this can be circumvented in a fairly realistic setup model called the common reference string (CRS) model. Here, a trusted party samples a random (possibly non-uniform) public string before the protocol. Other commonly studied setup models include the designated verifier [QRW19] and random oracle models [BR93]. In the quantum world, recent works [MY22a, ABKK23] have considered two-party protocols where the parties share different registers of an entangled quantum state. These are sometimes known to provide stronger guarantees than the CRS model [BKS23]. Recently, more complex yet localized quantum setups have also garnered attention in two-party protocols [AGL24, MNY24].

Unbounded Adversaries. Due to their widespread use cases, NIZKs satisfying various security properties have been explored. Of these, perhaps the most important characterization is in regards to whether security holds against an unbounded adversary, or only a computationally-bounded one. For instance, *soundness* typically only requires that an *efficient* malicious prover cannot convince the verifier of a *false* statement. Likewise, the standard notion of *zero-knowledge* requires that an *efficient* verifier learns nothing from the transcript. However, one can consider upgrades of these in the form of *statistical soundness* and *statistical zero-knowledge*, which provide security against an unbounded prover and an unbounded distinguisher (equivalently, the verifier in the NIZK case) respectively. Hence, many NIZKs in the literature satisfy these stronger properties [BFM88, FLS90, GO94, GOS06, SW14, BP15, CCH⁺19, PS19, WWW25].

Statistical Soundness or Zero-Knowledge?. Although statistical variants of soundness and zero-knowledge are achievable from standard assumptions, it is unlikely to obtain both simultaneously, except for a small subclass of NP languages [PS05]. This is the case even for interactive protocols [For87], with evidence that quantum protocols do not help either [MW18]. Consequently, one has to choose which of these two properties to give up in favor of the other, depending on the application. Typically, statistical zero-knowledge is preferred as soundness only needs to hold

during the online phase, while zero-knowledge can be compromised anytime in the future to leak sensitive information. Still, this is not always ideal. Consider a large corporation (prover) that might have various means to cut down the time to break certain computational assumptions, with the verifier being an average user. In this case, a computational bounded assumption on the verifier could be more reasonable. A statistically sound NIZK can also remove the need to refresh the CRS periodically in such a setting. However, this introduces other issues, such as the verifier being able to sell the data to a powerful rival corporation.

A Quantum Compromise. Such a tradeoff seems inevitable in the classical setting, but previous works have proposed a novel compromise using quantum resources, in the context of interactive zero-knowledge [HMNY22, BK23]. In particular, the prover and verifier engage in a statistically-sound quantum protocol, after which the verifier is asked to return a *certificate of deletion*. Then, an everlasting guarantee akin to statistical zero-knowledge holds, given that the certificate is verified to be valid. Note that this only makes sense in a quantum world due to the no-cloning theorem, as the verifier can retain a classical proof indefinitely.

Observe also that the requirement to delete the proof does not interfere with verification. This is because verification can be performed first without disturbing the state (by gentle measurement), and the proof can be deleted having served its purpose. Another interesting aspect is that the verifier can immediately verify and revoke the proof, which can then be checked. As a result, the parties need not maintain the state for long unlike other primitives with certified-deletion. This can be quite useful due to the difficulty of maintaining quantum states.

The aforementioned works then showed that such proofs can be obtained for all languages in QMA (the quantum generalization of NP) assuming only OWFs, or even weaker quantum-assumptions such as the existence of pseudo-random states [JLS18]. Naturally, one can consider the non-interactive analogue where the prover sends a single message to the verifier, and the verifier sends back a deletion certificate. While one might argue this is no better than a two-round protocol, the deployed applications wouldn't be slowed down by the second message, which could be batched and sent at a later point. Hence, at the expense of using quantum resources and relying on the ability to penalize a non-confirming verifier, there exists the possibility of obtaining statistically-sound NIZKs with an everlasting zero-knowledge guarantee. Despite the hope, and the construction of many advanced primitives with certified-everlasting forms of security (See Section 1.3), CE-NIZKs are yet to be realized. We explore this possibility in detail.

1.2 Contributions

1. *Definitions and Impossibility.* We present definitions of certified-everlasting NIZK proofs (CE-NIZKs) in the CRS and shared-EPR models to begin with. Then, we examine natural approaches to constructing CE-NIZKs in the CRS model. Specifically, we identify a class of protocols we call *deletion-resistant* CE-NIZKs. At a high level, these admit a deletion algorithm that splits the proof state into two registers, one that passes verification of deletion, and another that passes proof verification. We then rule out the existence of such protocols by showing that they imply NIZKs that are both statistically-sound and statistically zero-knowledge. We also argue that non-interactive counterparts of known interactive protocols fall into this category.
2. *Feasibility in the CRS Model.* As a result of this barrier, we provide a unique approach to constructing CE-NIZKs in the CRS model, where we employ two-levels of NIZK proofs that are generated in quantum-superposition. In order to prove security, we employ the two-

slot technique [NY90] and the OR proof strategy [FLS90], and reduce to a certified-deletion theorem of BB84 states [BK23]. This allows us to construct a CE-NIZK for NP based on black-box use of a quantum-secure NIZK proof for NP¹ (Definition 3.6), along with just post-quantum OWFs. We observe that quantum-secure NIZKs (different from post-quantum ones) are implied by LWE due to the construction of [WWW25]. Hence, we obtain our result based on LWE, matching the state-of-the-art assumption for post-quantum NIZK.

3. *Bypassing the Impossibility with Shared Entanglement.* We then explore a natural quantum setup model where the prover and verifier share entanglement in the form of halves of EPR pairs. Surprisingly, we find that the aforementioned impossibility no longer applies to this setting. As a consequence, we show that a generalization of the hidden-bits compiler of [QRW19] suffices to obtain CE-NIZKs for NP in this setting. In particular, we obtain CE-NIZKs based on any statistically binding hidden-bits generator [QRW19], which can be based on LWE [WWW25]. Although the assumption is the same as our CRS protocol, the protocol is highly efficient in regards to the quantum operations involved. For comparison, our CRS protocol generates and verifies proofs in superposition of a highly entangled state. On the other hand, this protocol only requires single-qubit measurements of the EPR halves in one of two basis, and uses classical computation and communication otherwise.

1.3 Related Work

Certified-Everlasting Cryptography. The pioneering work of Broadbent and Islam [BI20] constructed encryption with certified-deletion, where ciphertexts can be deleted in a way that the plaintext becomes unrecoverable (by an unbounded adversary) even if the secret-key is leaked later. This was followed by a series of works on certified-everlasting security, which refers to a bounded adversary that cannot break some guarantee in the future using unbounded computation, assuming it provides a valid certificate during the computationally bounded stage. We discuss these works briefly as follows.

The work of Hiroka, Morimae, Nishimaki and Yamakawa [HMNY22] constructed a certified-everlasting statistical-binding commitment scheme and used it to obtain a certified-everlasting zero-knowledge proof system for QMA. Later, the work of Bartusek and Khurana [BK23] introduced an elegant compiler for obtaining primitives with certified-deletion. Through it, they obtained PKE, ABE, FHE, commitments (and ZK via [HMNY22]), witness encryption, timed-release encryption, MPC protocols and secret-sharing schemes with certified-everlasting security. The work of Hiroka et al. [HKM⁺24] constructed public-key functional encryption (and other variants), compute-and-compare obfuscation and garbled circuits with such security guarantees. Advanced secret-sharing schemes with certified-everlasting security were studied in the work of Bartusek and Raizes [BR24]. The work of Champion, Kitagawa, Nishimaki and Yamakawa [CKNY25] constructed a certified-everlasting variant of untelegraphable encryption, a relaxation of unclonable encryption. A notion of differing inputs obfuscation (dIO) with certified-everlasting security was introduced and constructed in the work of Bartusek et al. [BGK⁺24]. Their construction provides the desirable property of publicly-verifiable deletion, which is studied in several works [KNY23, BKP23, BKM⁺23, KS25].

Quantum-Enabled NIZKs. In recent years, several works have studied NIZKs with interesting properties that are only feasible in a quantum world. However, these are quite different from our

¹We require two special notions from the NIZK: quantum-secure adaptive zero-knowledge and post-quantum non-adaptive zero-knowledge with statistical CRS indistinguishability which are satisfied by the LWE-based NIZKs of [Wat24, WWW25].

notion of certified-everlasting NIZK. In particular, many of these works construct schemes that prevent copying attacks, i.e., they ensure that the recipient of a proof cannot convince several other verifiers, without itself knowing a witness. These “unclonable” NIZKs and variants were studied in the concurrent works of Jawale and Khurana [JK24] and Goyal, Malavolta and Raizes [GMR24]. A related notion was explored by the work of Abbaszadeh and Katz [AK25] where the recipient must generate a valid proof of deletion, and cannot convince another verifier thereafter. While the unclonable variants were shown to imply public-key quantum money, this certified-deletion variant was obtained from standard assumptions.

The work of Çakan, Goyal and Raizes [ÇGR24] studied NIZK with a stronger form of certified-deletion called certified-deniability. Intuitively, this is a simulation-based notion that guarantees that the state after deletion of the proof could be computed without the proof. The work presents a feasibility for this in the random oracle model (ROM), with evidence against its existence in the plain model. Our simulation definition shares some similarity to theirs, while additionally requiring statistical indistinguishability. Crucially, their impossibility does not contradict our feasibility in the CRS model, as our simulator has the ability to simulate the CRS. In the certified-deniability context, this ability would be restricted, similar to the work’s restriction on programmability in the ROM.

Finally, the work of [GLR⁺25] studies a notion of one-time security for NIZK and presents a construction from iO and LWE. This notion allows an authority to distribute quantum proving tokens to a prover. Then, the prover can only convince a verifier of as many statements as the number of tokens received. This notion is shown to imply public-key quantum money which justifies the use of the strong assumption of iO. We emphasize that all of these quantum NIZK notions are orthogonal to that of certified-everlasting zero-knowledge which we study. For a detailed comparison of our work with some of these works, see Section 2.6.

Revocable Cryptography. Revocable cryptography is closely related to the notions discussed above in that it involves a deletion (revocation) phase, after which certain information/capability is lost. This was pioneered by the work of Unruh [Unr14] which introduced revocable time-released encryption. Revocable notions for encryption/signatures were also studied in [AMP25, MPY24], and are similar to the notions of certified-deletion in the sense that secret data is being deleted. On the other hand, a long line of works on secure software leasing [AL21, KNY21, BJL⁺21, CMP24] and secure key leasing [KN22, APV23, AKN⁺23, PWYZ24, AHH24, CGJL25, KMY25, KNP25b, KNP25a, KLYY25] consider the following high-level premise. A user is provided a quantum state that can be used to evaluate some software or cryptographic functionality. Later, the user can provide a deletion-certificate, after which it is guaranteed that they can no longer evaluate the functionality. We remark that almost all these schemes only consider a computationally-bounded second-stage (post-deletion) adversary. This restriction is necessary in several contexts like the leasing of decryption keys of a PKE scheme.

2 Technical Overview

2.1 Defining Certified-Everlasting NIZKs

We will begin by defining the syntax of a certified-everlasting non-interactive zero-knowledge proof (CE-NIZK) (Definition 4.1). A CE-NIZK for a language $\mathcal{L} \in \text{NP}$ is a tuple of four algorithms (Setup, P, V, Cert) described as follows. The classical algorithm Setup(1^λ) outputs a common reference string crs as usual. The quantum prover’s algorithm P on input the CRS crs, a statement $x \in \mathcal{L}$ and corresponding witness $\omega \in R_{\mathcal{L}}(x)$, outputs a quantum proof state σ , along with a (possibly

classical) quantum state ρ_P , which will later be used to validate returned proofs. The quantum verifier's algorithm V on input crs , x and proof σ outputs a decision bit b along with a quantum state ρ_V . The quantum certification algorithm Cert takes as input quantum states ρ_P, ρ_V and outputs $\top$ (accept) or $\perp$ (reject). The context is that the verifier would verify the validity of the proof σ to obtain ρ_V (which would be close in trace-distance due to completeness and gentle-measurement) and send it to the prover who checks that the proof is revoked by running $\text{Cert}(\rho_P, \rho_V)$. Note that when we refer to a CE-NIZK for NP, we mean one that is a CE-NIZK for an NP-complete language.

We require a CE-NIZK for NP to satisfy the notions of completeness, statistical soundness and computational zero-knowledge which must also hold for a standard NIZK proof for NP. In addition, it must also satisfy the notion of certified-everlasting zero-knowledge (CE-ZK). Intuitively, the notion guarantees that the view of any QPT malicious verifier V^* can be simulated in a statistically close manner with just the statement x , conditioned on the fact that certification is successful. More formally, for every QPT malicious verifier V^* , there must exist a QPT simulator Sim_{V^*} such that the following guarantee holds for every true statement x and corresponding witness ω :

$$\left[\begin{array}{l} \text{crs} \leftarrow \text{Setup}(1^\lambda) \\ (\sigma, \rho_P) \leftarrow P(\text{crs}, x, \omega) \\ \rho_{A \otimes B} \leftarrow V^*(\text{crs}, x, \sigma) \\ \text{If } \text{Cert}(\rho_P, \rho_A) = \top \text{ then } \tilde{\rho} := \rho_B. \\ \text{Else } \tilde{\rho} := \perp. \\ \rho' \leftarrow \text{Sim}_{V^*}(x) \end{array} \right] \\ \tilde{\rho} \approx_s \rho'$$

Let us now explain this notion. First, a crs is sampled using Setup and a real proof σ and prover state ρ_P are computed. Then, the proof σ is provided to the malicious verifier, who outputs a bi-partite state $\rho_{A \otimes B}$. Then, the notion sets a state $\tilde{\rho}$ to be ρ_B (the residual state on register B) only if the state ρ_A on register A passes certification by $\text{Cert}(\rho_P, \rho_A)$. If not, $\tilde{\rho}$ is simply set to $\perp$, thereby giving a distinguisher no additional information when the certification is unsuccessful. Then, we require that the simulator Sim_{V^*} on input x produces a simulated state ρ' that is close in trace distance to $\tilde{\rho}$ from the real execution. Note that our protocols will come with a simulator that makes black-box use of V^* , but we define a non-black-box one here for generality.

2.2 Non-Existence of Deletion-Resistant Protocols

Before we introduce our approach to realizing CE-NIZKs, we will discuss some challenges in achieving them. Specifically, we identify a natural class of CE-NIZKs that are impossible to construct. The intuition is that these come with a deletion algorithm that splits the proof state into two, the first which passes prover certification, and a second which passes NP verification. Consequently, this operation can be performed at the prover's end before sending the second part to the verifier. We observe that these imply NIZKs which are both statistically sound and statistically zero-knowledge which are unlikely to exist. We define this subclass of CE-NIZKs formally as follows.

Definition 2.1 (Deletion-Resistant CE-NIZKs). A non-interactive protocol with the syntax of CE-NIZK (Definition 4.1) is said to be a deletion-resistant CE-NIZK if it satisfies the CE-ZK property, along with the following properties. There exist QPT algorithms $V^*, \tilde{V}$ such that the following conditions hold for every $x \in \mathcal{L}$, $\omega \in R_{\mathcal{L}}(x)$ and $\tilde{x} \notin \mathcal{L}$:

$$\Pr \left[\begin{array}{l} \text{crs} \leftarrow \text{Setup}(1^\lambda) \\ \text{Cert}(\rho_P, \rho_A) \rightarrow \top : \\ (\sigma, \rho_P) \leftarrow P(\text{crs}, x, \omega) \\ \rho_{A \otimes B} \leftarrow V^*(\sigma) \end{array} \right] \geq 1/\text{poly}(\lambda) \quad (1)$$

$$\Pr \left[\begin{array}{l} \tilde{V}(\text{crs}, x, \rho_B) \rightarrow 0 \mid \text{Cert}(\rho_P, \rho_A) \rightarrow \top : \\ \text{crs} \leftarrow \text{Setup}(1^\lambda) \\ (\sigma, \rho_P) \leftarrow P(\text{crs}, x, \omega) \\ \rho_{A \otimes B} \leftarrow V^*(\sigma) \end{array} \right] \leq \text{negl}(\lambda) \quad (2)$$

$$\Pr_{\text{crs} \leftarrow \text{Setup}(1^\lambda)} \left[\exists \tilde{\sigma} : \tilde{V}(\text{crs}, \tilde{x}, \tilde{\sigma}) \rightarrow 1 \right] \leq \text{negl}(\lambda) \quad (3)$$

Theorem 2.2. *If there exists a deletion-resistant CE-NIZK for a language $\mathcal{L}$, then there exists a quantum NIZK for $\mathcal{L}$ satisfying statistical soundness and statistical zero-knowledge.*

Proof. Let Π be a deletion-resistant CE-NIZK for a language $\mathcal{L}$. This means it has algorithms $(\text{Setup}, P, V, \text{Cert})$. Furthermore, there exist QPT algorithms $V^*, \tilde{V}$ satisfying the conditions of the theorem by assumption. We now define the following prover $\tilde{P}$:

$\tilde{P}(\text{crs}, x, \omega)$:

1. Compute $(\sigma, \rho_P) \leftarrow P(\text{crs}, x, \omega)$.
2. Compute $\rho_{A \otimes B} \leftarrow V^*(\sigma)$.
3. If $\text{Cert}(\rho_P, \rho_A) = \top$, then output ρ_B . Otherwise, repeat from Step 1.

Observe now that the NIZK proof $\tilde{\Pi} := (\text{Setup}, \tilde{P}, \tilde{V})$ is statistically-sound due to the third condition of the theorem. Also, completeness follows from the fact that $\tilde{P}$ will obtain $\rho_{A \otimes B}$ such that $\text{Cert}(\rho_P, \rho_A) = \top$ with overwhelming probability in polynomial time (due to the first condition of the theorem), and the fact that $\tilde{V}$ accepts ρ_B from the second condition. Finally, observe that by viewing V^* as a malicious QPT verifier attacking the CE-ZK property of Π , the state ρ_B output by $\tilde{P}$ can be statistically simulated. This means that $\tilde{\Pi}$ satisfies statistical zero-knowledge. $\square$ $\square$

Remark 2.3. *We remark that an analogous implication follows for protocols with weaker certified-everlasting witness-privacy notions, such as a certified-everlasting variant of non-interactive witness-indistinguishable (NIWI) proofs.*

2.3 On Generalizations of Prior Work

We now recall previous works on CE-ZK proofs in the interactive setting by Hiroka et al. [HMNY22] and Bartusek and Khurana [BK23]. The work of Hiroka et al. constructed CE-ZK proofs for languages in QMA based on a new primitive they introduced called commitments with statistical binding and certified-everlasting hiding. The certified-everlasting hiding property refers to the fact that after deletion, even an unbounded receiver is unable to learn anything about the contents of the commitment. While such commitments were constructed in the random oracle model in their work, Bartusek and Khurana constructed them from any statistical binding commitment scheme in the plain model. Note that such statistically binding commitments can be based on OWFs [Nao91] and from potentially weaker quantum-cryptographic assumptions [MY22b, AQY22].

We now argue that these protocols do not generalize easily for constructing CE-NIZKs. At a high level, the CE-ZK proofs established by these works rely on the following template:

Template for CE-ZK Proofs:

- First, the prover sends some $n = \text{poly}(\lambda, |x|, |w|)$ commitments $\{\text{com}^i\}_{i \in [n]}$ to the verifier, generated using the certified-everlasting commitment scheme.
- Then, the verifier sends a randomly chosen opening set $S \subset [n]$ to the prover, along with deletion certificates $\{\text{cert}^i\}_{i \in [n] \setminus S}$ corresponding to the commitments that are not to be opened.
- Finally, the prover sends the openings of the commitments $\{\text{open}^i\}_{i \in S}$ for the set S . These openings are then sufficient for the verifier to determine the validity of the statement.

Note that while these works use classical certificates $\{\text{cert}^i\}_{i \in [n] \setminus S}$, the same ideas also apply if these certificates were quantum. The main idea here is that since the unopened commitments are deleted by the verifier, even an unbounded distinguisher cannot tell apart the real and simulated views. Naturally, one might expect that a Fiat-Shamir based approach [FS87, CCH⁺19, PS19] in the CRS model could be employed to turn these interactive protocols into non-interactive ones. However, we argue that such protocols would be deletion-resistant, and hence run into Theorem 2.2. To see this, consider a CE-NIZK $\Pi = (\text{Setup}, \text{P}, \text{V}, \text{Cert})$ with a similar template, where the prover P determines the opening set S by itself, by applying a hash function to some classical part of the generated proof. Since V would only need to check the commitments in S as part of the protocol, we can consider $\tilde{\text{V}} := \text{V}$ and a malicious QPT verifier V^* that splits the commitments in S and the commitments in $[n] \setminus S$ into two different registers. Consequently, the state $\rho_{\text{A} \otimes \text{B}}$ produced by V^* is such that ρ_{A} would pass certification while ρ_{B} would be accepted by V . Notice that both verification and certification could check the validity of S , but the classical information that is hashed to produce S can simply be cloned. Moreover, designing V, Cert to artificially deviate from the template doesn't help. For instance, V clearly cannot check all the commitments, since the proof system should satisfy computational zero-knowledge. Moreover, even if Cert requires all the commitments to be revoked, V^* can easily clone the contents of the opened commitments. We note that natural attempts at generalizing the hidden-bits paradigm [FLS90] to the CE-NIZK setting do not work, due to a similar argument. Despite this barrier, we show next that CE-NIZKs are indeed possible to construct in the CRS model from standard assumptions.

2.4 Feasibility in the CRS Model

The high-level idea of our construction is to utilize an unclonable quantum state that is entangled with NIZK proofs. The hope is that this prevents the splitting attack discussed in the previous section. We require several concrete ideas to realize this, which we present as follows. The main ingredient we require is a classical NIZK proof $\Pi_{\text{nizk}} = (\text{Setup}_{\text{nizk}}, \text{P}_{\text{nizk}}, \text{V}_{\text{nizk}})$ for NP. Our construction utilizes this NIZK scheme at two-levels, with an “inner” proof that is provided in classically encrypted form, and “outer” proofs that are entangled with a BB84 (Wiesner) state. The inner proof π_{in} is the actual NIZK proof, generated as $\pi_{\text{in}} \leftarrow \text{P}_{\text{nizk}}(\text{crs}_{\text{in}}, x, \omega)$ for a statement-witness pair (x, ω) and a CRS sampled as $\text{crs}_{\text{in}} \leftarrow \text{Setup}_{\text{nizk}}(1^\lambda)$. The problem is that if π_{in} were given out in the clear, then the protocol cannot satisfy CE-ZK as Π_{nizk} is statistically-sound. To remedy this, we sample ℓ -many BB84 quantum states of the form $|y^i\rangle^{\theta^i} := H^{\theta^i[1]} |y^i[1]\rangle \otimes \dots \otimes H^{\theta^i[\lambda]} |y^i[\lambda]\rangle$ where $y^i, \theta^i \leftarrow \{0, 1\}^\lambda$, $y^i[j]$ denotes the j -th bit of y^i , H is the Hadamard transform, and ℓ is the size of π_{in} . Consider now the $\ell\lambda$ -qubit BB84 state $|y\rangle_{\text{R}}^{\theta} = \sum_{\mathbf{z}} \alpha_{\mathbf{z}} |\mathbf{z}\rangle_{\text{R}}$ where $y := y^1 \parallel \dots \parallel y^\ell$ and $\theta^1 := \theta^1 \parallel \dots \parallel \theta^\ell$. Let us now define a function f and a value pad such that $\text{pad} := f(\theta, y) := \bigoplus_{j: \theta^1[j]=0} y^1[j] \parallel \dots \parallel \bigoplus_{j: \theta^\ell[j]=0} y^\ell[j]$. In other words, each of the ℓ -bits of pad are obtained by taking the XOR of the computational basis positions of the corresponding λ -size “block” of the BB84 state $|y\rangle_{\text{R}}^{\theta}$. Then, instead of π_{in} being provided to the adversary directly, it is one-time pad encrypted as $\pi_{\text{in}} \oplus \text{pad}$. The purpose of this one-time pad will be clear later, when we invoke a certified-deletion theorem of BB84 states by

[BK23] to ensure that π_{in} is information-theoretically removed from the verifier’s view. Note also that in actuality, we need to provide an OTP encryption $\text{ct} := \pi_{\text{in}} \oplus \text{pad} \oplus k$ for $k \leftarrow \{0, 1\}^\ell$ instead of just $\pi_{\text{in}} \oplus \text{pad}$ due to some technicalities in invoking the certified-deletion theorem of [BK23].

The important aspect here is that every term z in the superposition of the quantum state $|y\rangle_{\text{R}}^\theta$ can be used to compute the common value pad using θ as $\text{pad} = f(\theta, z)$. We leverage this fact to produce the “outer” proofs π_{out}^z in superposition as $|\psi\rangle_{\text{R}\otimes\text{P}} := \sum_z |z\rangle_{\text{R}} |\pi_{\text{out}}^z\rangle_{\text{P}}$. In particular, each π_{out}^z proves the following NP statement x^z :

$$\text{Statement } x^z : \exists(\theta, k) : V_{\text{nizk}}(\text{crs}_{\text{in}}, x, \text{ct} \oplus f(\theta, z) \oplus k) = 1$$

We observe that by proving the statement x^z , the proof π_{out}^z also proves the validity of the statement x . We argue this using the *statistical adaptive* soundness property of Π_{nizk} . First, observe that completeness follows from the fact that $\text{ct} \oplus f(\theta, z) \oplus k = \pi_{\text{in}}$ for each z in the superposition of $|\psi\rangle$, and $V_{\text{nizk}}(\text{crs}_{\text{in}}, x, \pi_{\text{in}}) = 1$ follows from the completeness of Π_{nizk} . Now, consider the case when x is false, which means there is no accepting proof $\tilde{\pi}_{\text{in}}$ that makes $V_{\text{nizk}}(\text{crs}_{\text{in}}, x, \cdot)$ accept (whp over choice of crs_{in}). As a result, all of the derived statements x^z would also be false and hence would be rejected whp by $V_{\text{nizk}}(\text{crs}_{\text{out}}, x^z, \cdot)$ where $\text{crs}_{\text{out}} \leftarrow \text{Setup}(1^\lambda)$ is another part of the full CRS $\text{crs} := \text{crs}_{\text{in}} \parallel \text{crs}_{\text{out}}$. Observe that if Π_{nizk} were only computationally sound (which we don’t consider), then such an argument doesn’t make sense as there would exist accepting proofs for false statements as well. Notice also that adaptive soundness ensures that even if the statements x^z are based on crs_{out} , soundness still holds. Furthermore, computational zero-knowledge follows due to the fact that $(\text{crs}_{\text{in}}, \pi_{\text{in}})$ can be replaced with $(\text{crs}_{\text{Sim}}, \pi_{\text{Sim}})$ which are guaranteed by the zero-knowledge of Π_{nizk} . Let us now focus on the certified-everlasting zero-knowledge (CE-ZK) property.

The rough idea is to reduce to the aforementioned deletion theorem for BB84 states by Bartusek and Khurana. Intuitively, the theorem allows us to replace each bit of the plaintext π_{in} inside the OTP encrypted ciphertext ct with a bit of our choice in a statistically indistinguishable way, conditioned on the fact that the BB84 state $|y\rangle^\theta$ is verifiably returned. In order to enforce such a revocation, we require the verifier send back the original state $|\psi\rangle_{\text{R}\otimes\text{P}}$ and have the prover uncompute the proofs on the P register, followed by verifying if the resulting state is equivalent to $|y\rangle^\theta$ or not. However, there is an important detail involved in invoking their deletion theorem. It only guarantees security if the state provided to the adversary computationally hides the basis information θ . However, the problem with the real proofs π_{out}^z is that they require (θ, k) as the witness. Fortunately, this is exactly what the zero-knowledge guarantee of Π_{nizk} for the outer proofs provides us, allowing to simulate the proof state without access to (θ, k) . However, the traditional notion of computational zero-knowledge is insufficient, as the outer proofs are generated for a superposition of statements wrt the same CRS. We observe that such a quantum-secure zero-knowledge notion (first explored in [ACE⁺24]) is satisfied by the LWE-based NIZKs of [Wat24, WWW25]. Apart from this quantum-secure ZK property, we need one more special property from the underlying NIZK. In particular, we require that the marginal distribution of the CRS output by the non-adaptive ZK simulator of Π_{nizk} as $(\text{crs}_{\text{Sim}}, \pi_{\text{Sim}}) \leftarrow \text{Sim}_{\text{nizk}}(x)$, is statistically close to the real CRS distribution. Once again, we find that NIZKs obtained via the hidden-bits paradigm such as the LWE ones mentioned above satisfy this property. On the other hand, we find that LWE-based NIZK proofs obtained via the Fiat-Shamir approach [CCH⁺19, PS19] do not satisfy this property as their simulator samples a “fake” key for a keyed correlation-intractable hash function, that is only computationally indistinguishable from a real key. It is also unclear if these Fiat-Shamir NIZKs satisfy quantum-secure ZK. Now, we explain how CE-ZK is proved using these properties.

The main idea is to swap out the plaintext π_{in} bit-by-bit into a simulated proof π_{Sim} generated by $(\text{crs}_{\text{Sim}}, \pi_{\text{Sim}}) \leftarrow \text{Sim}_{\text{nizk}}(x)$, in a statistically indistinguishably way using the deletion theorem. Note

that we would first switch crs_{in} into crs_{Sim} due to the aforementioned statistical CRS indistinguishability. The problem with this argument is that the moment a bit of π_{in} is flipped to get π'_{in} , there is no guarantee that the computational zero-knowledge of the proofs π_{out}^z (which are generated using the witness $\omega_{\text{out}} := (\theta, k)$) holds as π'_{in} need not be accepted by $V_{\text{nizk}}(\text{crs}_{\text{Sim}}, x, \cdot)$. Consequently, we employ the two-slot technique [NY90], which was also used by [BGK⁺24] in an iO-centric certified-deletion context. In more detail, we utilize a $2\ell\lambda$ qubit BB84 state $|y\rangle^\theta$ which encodes two pads $\text{pad}^0, \text{pad}^1$ and provide two ciphertexts $\text{ct}^0 := \pi_{\text{in}} \oplus \text{pad}^0 \oplus k^0$ and $\text{ct}^1 := 0^\ell \oplus \text{pad}^1 \oplus k^1$. Moreover, the statements x^z corresponding to the superposition terms of the BB84 state are modified as follows:

$$\begin{aligned} \text{Statement } x^z : \exists(\theta, k^0, k^1) : V_{\text{nizk}}(\text{crs}_{\text{in}}, x, \text{ct}^0 \oplus f^0(\theta, z) \oplus k^0) = 1 \\ \text{OR} \\ V_{\text{nizk}}(\text{crs}_{\text{in}}, x, \text{ct}^1 \oplus f^1(\theta, z) \oplus k^1) = 1 \end{aligned}$$

where f^0 is defined similarly to f for the last ℓ -qubits (which encode pad^0) while f^1 corresponds to the first ℓ -qubits (which encode pad^1). Due to the nature of this OR proof, we are able to swap the 0^ℓ plaintext in ct^1 into a simulated proof π_{Sim} one bit at a time. This means we no longer have the aforementioned issue as π_{in} inside ct^0 is sufficient to satisfy the OR proof, and hence invoke quantum-secure zero-knowledge of the outer proofs. Then, once π_{Sim} is placed in ct^1 , we can replace π_{in} bit-by-bit into 0^ℓ since π_{Sim} also satisfies $V_{\text{nizk}}(\text{crs}_{\text{Sim}}, x, \pi_{\text{Sim}}) = 1$ by zero-knowledge. Finally, observe that the resulting distribution can be produced with the statement x alone, ensuring that the CE-ZK property is satisfied. Note that apart from the NIZK for NP, we also rely on a quantum-secure PRF (implied by OWFs) to simulate uniform randomness in superposition, used to compute and uncompute the NIZK proofs. We also face the following subtle issue in the reduction to the deletion theorem of BB84 states. The quantum state returned by the adversary may also have superposition terms $\tilde{z}$ such that (θ, k^0, k^1) is not a valid witness for statement $x^{\tilde{z}}$. Consequently, we cannot argue that the state after uncomputation of real proofs is computationally close to the state after uncomputation of simulated proofs, based on the quantum-secure adaptive zero-knowledge property. To circumvent this issue, we employ a simpler (information-theoretic) variant of the key-testability technique of [KNY23, KNP25b]. At a high level, this involves producing an information-theoretic MAC tag for each term of the BB84 state, which can be verified after revocation. Intuitively, this ensures that the adversary cannot produce a state with different superposition terms (that have non-negligible amplitude) without breaking unforgeability of the MAC. In summary, we obtain the following theorem:

Theorem 2.4 (CE-NIZK in the CRS Model). *There exist CE-NIZKs for NP in the CRS model satisfying 1) statistical soundness, 2) computational zero-knowledge, and 3) certified-everlasting zero-knowledge, assuming the existence of OWFs and NIZKs for NP in the CRS model satisfying 1) statistical (adaptive) soundness, 2) computational (non-adaptive) zero-knowledge with statistical CRS indistinguishability, and 3) quantum-secure (adaptive) zero-knowledge.*

Since such NIZKs for NP are known from LWE due to [WWW25], we have:

Corollary 2.5 (CE-NIZK from LWE). *There exist CE-NIZKs for NP in the CRS model based on the polynomial hardness of the LWE problem.*

2.5 Quantum-Efficient Protocol in the Shared EPR Model

Previously, we showed that generalizations of the Fiat-Shamir and hidden-bits paradigms to the CE-NIZK setting do not work, as they run into Theorem 2.2. We observe that this impossibility does

not apply to the shared EPR model where the parties are initialized with halves of polynomially-many EPR pairs. Such a model has recently garnered attention in the context of non-interactive MPC [ABKK23, BKS23] and NIZK for QMA [BKS23, MY22a]. Recall that Theorem 2.2 relies on the following idea: a prover could split the quantum proof the verifier is supposed to receive into two parts, one which passes certification while the other suffices for verification. Hence, the prover need only send the part which is required for verification. This then implies a statistically secure NIZK. In the EPR model, the prover cannot perform this “split” as it does not control the entire quantum proof state, since the verifier’s EPR halves do not originate from the prover.

Our observation has some similarities to the work of [BKS23], which separated the CRS and shared EPR models in the context of non-interactive randomized oblivious transfer. In our case, the separation between the models only applies to the aforementioned deletion-resistant protocols, as evident from our CRS feasibility result. However, we demonstrate that this difference is useful as well, by constructing a protocol in the shared EPR model that is much more efficient in its use of quantum resources. Specifically, our protocol only involves performing single-qubit computational or Hadamard basis measurements on the EPR halves, and classical computation otherwise. We proceed to describe our approach.

We make use of the well-known hidden-bits paradigm [FLS90] that we adapt to the shared EPR model for the context of obtaining certified-everlasting ZK. We note that Fiat-Shamir variants of previous interactive protocols would also bypass the impossibility in the EPR model. However, we find the hidden-bits approach to be cleaner and also helps to avoid the statistically far simulated CRS issue with [CCH⁺19, PS19] mentioned earlier. We also emphasize that we assume a common reference string in addition to the shared EPRs. This is because it is not clear how to simulate a structured reference string using EPRs unlike a common random string. Moreover, statistically sound NIZKs from LWE [CCH⁺19, PS19, Wat24, WWW25] are all in the common reference string model. While we could consider a more complex entangled state that could simulated arbitrary reference strings, we stick with EPR setups for their simplicity and practicality. We believe that the NIZK proof of Bitansky and Paneth [BP15] based on iO and OWFs would also work for our purpose, allowing to use EPRs alone as it only requires a uniform CRS.

The hidden bits paradigm for constructing NIZK utilizes two components: (1) a hidden bits generator (HBG) and (2) a statistical sound and statistically zero-knowledge NIZK in the hidden-bits model. In the (idealized) hidden-bits model, there is a uniformly random string that is sampled and made available to the prover, but is hidden from the verifier. Then, the prover has the ability to specify some indices of its choice to a trusted third party, which then reveals the corresponding hidden bits to the verifier (and nothing about the other bits). Since NIZKs in the hidden-bits model are known unconditionally [FLS90], most works focus on designing an HBG. Intuitively, the purpose of the HBG is to realize the idealized hidden bits in the CRS model. In particular, it allows the prover P to sample a succinct commitment com , which along with the CRS, fixes a hidden bit string. The HBG then allows the prover to provably open certain positions of this fixed hidden bit string to the verifier. Crucially, the *statistical-binding* security of the HBG ensures that the prover is bound to this string, and cannot reveal anything else to the verifier. Moreover, *computational hiding* ensures that all unopened hidden bits are pseudorandom in the verifier’s view.

Our idea is that instead of determining the actual hidden bits string via the commitment com and the CRS crs , these are utilized to fix a common basis string θ for both parties. Then, the prover would reveal parts of θ to the verifier, and both parties would obtain the actual hidden bits by measuring their halves of the EPR pairs in the θ basis. Specifically, the hidden bits are set to be the XOR of outcomes of computational basis positions corresponding to different “blocks” of the many EPR pairs. Note that the nature of EPR pairs ensures that both parties would derive the same measurement outcomes on measuring in the same basis. Also, statistical soundness is ensured

due to the observation that we can consider an equivalent verifier that measures all its EPRs in the computational basis before the prover is even initialized. Consequently, the prover can only manipulate the hidden bits via the basis string θ which is restricted by the HBG commitment com . Note that the inability of this hypothetical verifier to pass certification is not a problem, as it is only used as a proof tool to argue soundness. While computational zero-knowledge follows easily by arguments similar to prior work [QRW19], certified-everlasting zero-knowledge is argued based on the following fact. The verifier is asked to delete all the unopened hidden bits by measuring the corresponding EPRs in the Hadamard basis. Consequently, even if computational hiding of the HBG eventually leaks θ to an unbounded distinguisher, the certified-deletion theorem of [BK23] can be applied to ensure that these hidden bits are information-theoretically hidden. Consequently, we have:

Theorem 2.6. *CE-NIZKs for NP exist in the shared EPR Model (Definition 4.3), assuming statistical binding hidden bits generators (Definition 6.1) exist.*

From the fact that statistical binding HBGs are known from LWE [WWW25], and due to the structure of our protocol, we have the following corollary:

Corollary 2.7. *Assuming the polynomial hardness of LWE, there exist CE-NIZKs for NP in the shared EPR Model, where the prover and verifier only perform single-qubit measurements apart from arbitrary classical computation.*

2.6 Relation with Other Quantum-Enabled NIZKs

The works of [JK24] and [GMR24] introduced unclonable NIZK which is primitive that at a high level, guarantees the following property. Consider an adversary that is provided with a quantum proof of some NP statement x . Then, if it can produce two separate quantum proofs that both pass verification, then one can efficiently extract out a witness w for the statement x from the adversary. Intuitively, this ensures that if an adversary only has the proof and not the witness, it cannot produce more than one valid proof. Clearly, this is only feasible in the quantum setting due to the no-cloning principle. In [JK24], it was also informally mentioned that such proofs are a prerequisite for obtaining NIZKs with some form of certified deletion, while also mentioning certified everlasting security. The intuition is that if the adversary could clone proof states, then it could delete one of them successfully while retaining the other. Moreover, these works show that unclonable NIZK implies public-key quantum money (PKQM). On the other hand, we establish CE-NIZK in this paper from LWE, which is not known to imply PKQM. The reasoning is that CE-NIZKs need not be unclonable in the sense of [JK24], which actually doesn't require the adversary to generate two copies of the state. To see this, observe the following attack on our CE-NIZK in the CRS model that clearly breaks unclonable security. The adversary on obtaining the proof state $|\psi\rangle_{\text{R}\otimes\text{P}} = \sum_{\mathbf{t}} |\mathbf{t}\rangle$, simply applies CNOTs to get the state $\sum_{\mathbf{t}} |\mathbf{t}\rangle |\mathbf{t}\rangle$ and gives the two entangled registers to two verifiers. Since verification is done entirely in the computational basis, both verifiers would accept the proof. We also mention that by utilizing coset states along with iO (which implies PKQM assuming OWFs) instead of BB84 states, one might be able to obtain both unclonability and certified-everlasting security.

In the related work of [AK25], a NIZK with certified deletion (NIZK-CD) was obtained from standard assumptions, specifically from any NIZK (with certain properties) and a statistically binding commitment scheme. The security notion intuitively ensures that if an adversary is able to pass deletion and simultaneously produce a valid proof, then one can efficiently extract a witness from the adversary. Our construction also seems to at-least intuitively ensure that an adversary

that deletes cannot produce a valid proof. Consequently, it might be desirable to formally explore the possibility of a NIZK satisfying both everlasting security and the inability to prove, with a single deletion certificate. We emphasize that it is unclear whether CE-NIZK implies NIZK-CD. This is because it might be possible for an adversary to delete the proof of a CE-NIZK scheme, while somehow retain information in the form of a different proof that satisfies statistical zero-knowledge. While this doesn't contradict certified-everlasting security, it still allows the adversary to prove.

2.7 Open Questions

1. *Publicly-Verifiable Deletion:* While our construction only satisfies privately-verifiable deletion, obtaining the publicly-verifiable variant is more desirable. We believe this should be possible by employing coset states [CLLZ21] instead of BB84 states, along with iO (by employing techniques from [BGK⁺24]). Note that coset states with iO and OWFs are sufficient to obtain public-key quantum money (PKQM). However, it is not clear that CE-NIZK with publicly-verifiable deletion implies PKQM because an efficient PKQM verifier may be unable to exploit the fact that an adversarial state contains some information about the witness. Still, it is not obvious how to obtain publicly-verifiable deletion without iO, and we leave it as an interesting open question.
2. *Classical Certificates:* Our construction in the CRS model requires quantum revocation, while it is ideal if the verifier could only send back a classical certificate. One approach towards achieving this would be to utilize a hybrid quantum fully homomorphic encryption (QFHE) scheme that is known from LWE [Mah20, Bra18], as used in the works of [Shm22a, Shm22b]. In particular, the verifier could be provided with a QFHE ciphertext that encrypts (θ, k^0, k^1) so that it could perform the uncomputation of the superposition proofs using QFHE evaluation. This way, it would be able to obtain a certificate that is a QFHE encryption of Hadamard basis measurements, which could be decrypted and checked by the prover. The problem here is that we cannot simply replace the QFHE plaintext with a dummy, followed by using a similar certified-deletion argument as [BK23]. This is because deletion verification requires the QFHE secret-key, making it unclear how to reduce to the semantic security of QFHE. Despite this, there might be new techniques that would allow to prove the security of such a construction, and we leave this for future work.
3. *Certified-Everlasting NIWIs in the Plain Model:* It is also interesting to consider other non-interactive proof systems such as NIWIs, which are possible to obtain even in the plain model [DN00, BOV03]. In the post-quantum setting, these can be based on post-quantum iO [BP15]. In particular, one can try to obtain a NIWI with statistical soundness along with computational witness indistinguishability (WI) and certified-everlasting WI in the plain model. Our construction seems to be general enough to work for this case, except that the outer proofs generated in superposition can no longer be simulated by the computational zero-knowledge property. Recall that this is an important part of our proof, as it is needed to invoke the certified deletion theorem. It might still be possible just to rely on the witness indistinguishability property. However, we think this approach would require a stronger deletion theorem, and possibly techniques from leakage-resilient cryptography [KR19, QGLZR24].
4. *CE-NIZKs for QMA:* In this work, we focus on obtaining CE-NIZK for NP, due to the challenges in obtaining NIZK for QMA in the CRS model (See [Shm21]). On the other hand, interactive CE-ZK is known for all of QMA [HMNY22]. In this regard, it would be interesting to explore CE-NIZK for QMA in other setup models. For inspiration, the works

of [Shm21] and [MY22a] show NIZKs for QMA in the designated-verifier and shared-EPR models respectively, by utilizing variants of the Fiat-Shamir paradigm. On the other hand, one could consider CRS model constructions that make heuristic use of iO [BJRT25].

3 Preliminaries

3.1 Notation

We denote a (possibly quantum) algorithm P with input 1^λ and outputs $\mathbf{m}, \mathbf{ct}$ as $P(1^\lambda) \rightarrow (\mathbf{m}, \mathbf{ct})$ or $(\mathbf{m}, \mathbf{ct}) \leftarrow P(1^\lambda)$. Here, λ denotes a security parameter and $1^\lambda := 1 \parallel \dots (\lambda \text{ times}) \parallel 1$, where $\parallel$ denotes concatenation and $:=$ denotes defining (also used for assigning/substituting). We use $P(1^\lambda)[1]$ and $P(1^\lambda)[2]$ to denote the first and second indices of the output tuple respectively. For a set S , by $s \leftarrow S$, we mean s to be sampled uniformly from S . For a distribution D , by $s \leftarrow D$, we mean sampling s according to D . The functions $\text{poly}(\lambda)$ and $\text{negl}(\lambda)$ denote a polynomial and a negligible function in λ respectively. We use the notation $\mathcal{A}^{P(\text{crs}, \cdot, \cdot)}$ to denote that the quantum algorithm $\mathcal{A}$ gets quantum access to a quantum algorithm $P(\text{crs}, \cdot, \cdot)$ (with classical inputs and outputs) on the second and third inputs. For a language $\mathcal{L} \in \text{NP}$, we consider the relation $R_{\mathcal{L}}$ consisting of statement-witness pairs. Hence, $R_{\mathcal{L}}(x)$ denotes the set of witnesses of a statement x . We denote quantum registers as A, B, R etc. We use the notation y_i to denote the i -th bit of a string y , and R_i to denote the i -th qubit of R . Sometimes, we also use $y[i]$ to denote the i -th bit. Additionally, we use the notation y_I to denote $\{y_i\}_{i \in I}$ for some set I . Note that y^i doesn't mean the i -th bit of y . However, the notation y^I denotes $\{y^i\}_{i \in I}$. Note also the subscripts of Sim_1 and π_{nizk} do not correspond to indexing, as the first refers to an algorithm and the subscript of the latter is in a serif font. Sometimes, we use $\bar{I}$ to denote the complement of a set I , when the universe is clear from context. We also employ the notation $s_I := t_I \oplus r_I$ to denote $s_i := t_i \oplus r_i$ for each $i \in I$. For a classical algorithm C , we use $C(s; r)$ to denote running it on input s and random tape r . For two distributions D_0, D_1 , by the notation $D_0 \approx_c D_1$, we mean the two are computationally indistinguishable by quantum polynomial time (QPT) adversaries with non-uniform quantum advice. Often, we simply use $d_0 \approx_c d_1$ for samples $d^0 \leftarrow D_0, d^1 \leftarrow D_1$ drawn from these distributions to denote the same. We use $D_0 \equiv D_1$ to denote the distributions are identical and $D_0 \approx_s D_1$ to denote they are statistically close.

3.2 Quantum Information

A pure state $|\psi\rangle$ is a vector in a complex Hilbert space $\mathcal{H}$ with unit norm $\| |\psi\rangle \| = 1$. An operator P which satisfies $P^\dagger = P$ is said to be a Hermitian operator. Let $S(\mathcal{H})$ be the set of Hermitian operators on $\mathcal{H}$. A density matrix $\rho \in S(\mathcal{H})$ is a positive semi-definite operator with unit trace $\text{Tr}(\rho) = 1$. A density matrix represents a mixture over pure states, and the density matrix of a pure state $|\psi\rangle$ is $|\psi\rangle\langle\psi|$. A Hilbert space can be split into registers $\mathcal{H} := \mathcal{H}_{R_1} \otimes \mathcal{H}_{R_2}$. A unitary matrix U is one that satisfies $UU^\dagger = U^\dagger U = \mathbb{I}$. Applying U to a state $|\psi\rangle$ transforms it into the state $U|\psi\rangle$. For a mixed state ρ , applying U transforms it into $U\rho U^\dagger$. A projector Π is a Hermitian operator additionally satisfying $\Pi^2 = \Pi$. The trace distance between two mixed states ρ, σ is given by $\text{TD}(\rho, \sigma) := \frac{1}{2} \text{Tr}(\sqrt{(\rho - \sigma)^\dagger (\rho - \sigma)})$. It provides an upper bound on the distinguishing advantage of ρ, σ by any quantum algorithm. For quantum states ρ and σ , $\rho \approx_s \sigma$ denotes that the states are negligibly close in trace distance. A quantum polynomial-time (QPT) algorithm refers to a polynomial-size quantum circuit composed of some universal gate set, along with some non-uniform advice state $|\psi\rangle$. We denote the Hadamard transform/operator by H , which is defined

as $H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. For $y, \theta \in \{0, 1\}^n$ for some polynomial n in λ , we consider BB84 (Wiesner) states of the following form, where y_i, θ_i denote the i -th bits of y, θ respectively:

$$|y\rangle^\theta := H^{\theta_1} |y_1\rangle \otimes \dots \otimes H^{\theta_n} |y_n\rangle$$

We denote pure quantum states as $|\psi\rangle, |z\rangle$ etc while the quantum states ρ, σ will either refer to pure or mixed states based on the context. We use the notation $|\psi\rangle_{\text{R}}$ to denote a state on register R and $\rho_{\text{A} \otimes \text{B}}$ to denote a bi-partite state on register $\text{A} \otimes \text{B}$. We denote the state on B obtained by tracing out A as $\rho_{\text{B}} := \text{Tr}_{\text{A}}(\rho_{\text{A} \otimes \text{B}})$. For simplicity, we often use ρ_{B} to denote the state on register B at different points when it is clear from context. For e.g., we do not use a different ρ'_{B} to refer to the state obtained after A is collapsed to some specific value. An EPR pair is the following state on a pair of registers P, V: $\frac{1}{\sqrt{2}}(|0\rangle_{\text{P}}|0\rangle_{\text{V}} + |1\rangle_{\text{P}}|1\rangle_{\text{V}})$, where P, V are often referred to as EPR halves.

3.3 Cryptography Basics

Definition 3.1 (Learning with Errors (LWE) [Reg05]). Consider integer functions $n, m, q \in \mathbb{Z}$ of the security parameter λ . Let $\chi(\lambda)$ be an error distribution over $\mathbb{Z}$. The LWE assumption is said to hold if for every QPT adversary $\mathcal{A}$, there exists a negligible function $n(\lambda)$ such that:

$$\left| \Pr \left[\mathcal{A}(A, y) \rightarrow 1 \mid \begin{array}{l} A \leftarrow \mathbb{Z}_q^{n \times m} \\ s \leftarrow \mathbb{Z}_q^n \\ e \leftarrow \chi^m \\ y := s^\top A + e \end{array} \right] - \Pr \left[\mathcal{A}(A, u) \rightarrow 1 \mid \begin{array}{l} A \leftarrow \mathbb{Z}_q^{n \times m} \\ u \leftarrow \mathbb{Z}_q^m \end{array} \right] \right| \leq n(\lambda)$$

Definition 3.2 (Quantum-Secure Pseudo-Random Function). Let $\{\text{F}_{\text{prfk}} : \{0, 1\}^{\ell_{\text{in}}} \rightarrow \{0, 1\}^{\ell_{\text{out}}}\}_{\text{prfk} \in \{0, 1\}^\lambda}$ be an efficiently computable family of functions, where $\ell_{\text{in}}, \ell_{\text{out}}$ are some polynomials in λ . Then, $\{\text{F}_{\text{prfk}}\}_{\text{prfk}}$ is said to be a quantum-secure PRF if for every QPT adversary $\mathcal{A}$, there exists a negligible function $n(\lambda)$ such that the following holds, where $\mathcal{R}$ is the set of all functions with input ℓ_{in} and output ℓ_{out} :

$$\left| \Pr \left[1 \leftarrow \mathcal{A}^{f(\cdot)}(1^\lambda) \mid f \leftarrow \mathcal{R} \right] - \Pr \left[1 \leftarrow \mathcal{A}^{\text{F}_{\text{prfk}}(\cdot)}(1^\lambda) \mid \text{prfk} \leftarrow \{0, 1\}^\lambda \right] \right| \leq n(\lambda)$$

Theorem 3.3 ([Zha12]). *Quantum-secure PRFs exist, assuming post-quantum secure one-way functions exist.*

Lemma 3.4 (One-Way to Hiding (O2H) [AHU19]). *Consider any functions $G, H : \mathcal{X} \rightarrow \mathcal{Y}$, an arbitrary value z , and some set $S \subseteq \mathcal{X}$ such that for every $x \notin S$, $G(x) = H(x)$. The values G, H, S, z may be arbitrarily distributed. Consider now an oracle algorithm $\mathcal{A}$ that makes at most q queries to its oracle. Let $\mathcal{B}$ be an oracle algorithm such that $\mathcal{B}^H$ executes as follows on input z :*

$\mathcal{B}^H(z)$:

1. Choose $i \leftarrow [q]$.
2. Execute $\mathcal{A}^H(z)$ until it makes its i -th query.
3. Measure the i -th query and output the outcome.

Then, the following holds:

$$\left| \Pr [\mathcal{A}^H(z) = 1] - \Pr [\mathcal{A}^G(z) = 1] \right| \leq 2q \cdot \sqrt{\Pr [\mathcal{B}^H(z) \in S]}$$

Remark 3.5. It is easy to see from the proof of [AHU19] that the lemma holds even when the auxiliary input z is some quantum state ρ_z . A similar observation was made in [HXY19], where it was mentioned that z could even be exponentially large, and hence allows to describe quantum states as well.

3.4 Quantum-Secure Non-Interactive Zero-Knowledge

Definition 3.6. A triple of PPT algorithms (Setup, P, V) is a quantum-secure non-interactive zero-knowledge proof (NIZK) in the common reference string (CRS) model for a language $\mathcal{L}$, if it satisfies the following properties:

- *Completeness:* The following holds for every $x \in \mathcal{L}$ and $\omega \in R_{\mathcal{L}}(x)$:

$$\Pr \left[V(\text{crs}, x, \pi) \rightarrow 0 : \begin{array}{l} \text{crs} \leftarrow \text{Setup}(1^\lambda) \\ \pi \leftarrow P(\text{crs}, x, \omega) \end{array} \right] \leq \text{negl}(\lambda)$$

- *Statistical (Non-Adaptive) Soundness:* For every $\tilde{x} \notin \mathcal{L}$, the following holds:

$$\Pr_{\text{crs} \leftarrow \text{Setup}(1^\lambda)} [\exists \tilde{\pi} : V(\text{crs}, \tilde{x}, \tilde{\pi}) \rightarrow 1] \leq \text{negl}(\lambda).$$

- *Statistical Adaptive Soundness:*

$$\Pr_{\text{crs} \leftarrow \text{Setup}(1^\lambda)} [\exists (\tilde{x}, \tilde{\pi}) : \tilde{x} \notin \mathcal{L} \wedge V(\text{crs}, \tilde{x}, \tilde{\pi}) \rightarrow 1] \leq \text{negl}(\lambda).$$

- *Computational Zero-Knowledge with Statistical CRS Indistinguishability:* There exists a QPT simulator Sim such that for all $x \in \mathcal{L}$ and $\omega \in R_{\mathcal{L}}(x)$, the following holds wrt non-uniform QPT distinguishers:

$$\left[\begin{array}{l} (x, \text{crs}, \pi) \approx_c (x, \text{crs}', \pi') \\ \wedge \\ (x, \text{crs}) \approx_s (x, \text{crs}') \end{array} : \begin{array}{l} \text{crs} \leftarrow \text{Setup}(1^\lambda) \\ \pi \leftarrow P(\text{crs}, x, \omega) \\ (\text{crs}', \pi') \leftarrow \text{Sim}(x) \end{array} \right]$$

Remark 3.7. Statistical CRS indistinguishability cannot hold for the CRS output by S_1 of an adaptive ZK simulator (S_1, S_2) of a statistically-sound NIZK. This was mentioned in [Gro06], and can be seen from the proof of [Pas13].

- *Quantum-Secure (Adaptive) Zero-Knowledge:* There exists a QPT two-part simulator (S_1, S_2) such that the following holds for every QPT adversary $\mathcal{A}$ that only queries *valid* statement-witness pairs in superposition, i.e., terms of the form $|x, \omega\rangle$ where $x \in \mathcal{L} \wedge \omega \in R_{\mathcal{L}}(x)$:

$$\left[\begin{array}{l} \mathcal{A}^{S_2(\text{td}, \cdot)}(\text{crs}') \\ \approx_c \\ \mathcal{A}^{P(\text{crs}, \cdot, \cdot)}(\text{crs}) \end{array} : \begin{array}{l} \text{crs} \leftarrow \text{Setup}(1^\lambda) \\ (\text{crs}', \text{td}) \leftarrow S_1(1^\lambda) \end{array} \right]$$

Theorem 3.8 ([QRW19, WWW25, ACE⁺24]). *Assuming the polynomial hardness of the LWE problem, there exist NIZKs for NP in the CRS model with the following properties:*

- *Statistical (Adaptive) Soundness*
- *Quantum-Secure (Adaptive) Zero-Knowledge*
- *Computational (Non-Adaptive) ZK with Statistical CRS Indistinguishability*

In more detail, the work of [QRW19] showed that any statistical-binding hidden bits generator (HBG) (Definition 6.1) implies a statistically-sound NIZK. The works of [Wat24, WWW25] constructed such HBGs from LWE with sub-exponential and polynomial modulus respectively. In the work of [ACE⁺24], it was argued that the NIZK of [BP15] from iO and OWFs satisfies the stronger notion of quantum-secure (adaptive) zero-knowledge. We observe that their reasoning also applies to the NIZK of [WWW25] and the compiler of [QRW19] due to the following reason:

The high level intuition is that these constructions simulate the idealized hidden bits model (HBM) [FLS90] in the CRS model. Note that the idealized HBM provides statistical zero-knowledge (along with statistical soundness), and hence also quantum-secure zero-knowledge. The observation made by [ACE⁺24] is that the HBM to CRS model transformation also plays well with superposition queries. This is because the proof of the zero-knowledge property only requires switching the CRS distribution computationally, before appealing to the statistical security of NIZK in the HBM. Since the CRS is classical and independent of the statement-witness queries, the quantum-security implication follows. For more details, see Appendix D of [ACE⁺24].

Finally, we observe that in the HBG to NIZK compiler of [QRW19], the CRS in the construction is computed as $(\text{crs}_{\text{bg}}, s)$ where the former is the CRS of the HBG and s is a truly random string. In the simulation $\text{Sim}(x)$, crs_{bg} is computed the same way, but s is computed as $s := r_{\text{bg}} \oplus r_{\text{hb}}$ where r_{bg} is a pseudo-random string generated by the HBG, while r_{hb} is a close to uniform hidden-bit string output by the hidden bits simulator $\text{Sim}_{\text{hb}}(x)$. Hence, statistical CRS indistinguishability follows by the statistical zero-knowledge guarantee of the hidden bits NIZK proof. This can also be seen from the proof of computational zero-knowledge in our shared EPR construction in Section 6.2, which is similar to the one in [QRW19].

3.5 Certified Deletion of BB84 States

Theorem 3.9 (Certified Deletion of BB84 States [BK23]). *Consider a distribution $\mathcal{Z}(\cdot, \cdot, \cdot)$ with three arguments: 1) a λ -bit string θ , 2) a bit b , and 3) a λ -qubit register $\mathcal{D}$. For all $\theta \in \{0, 1\}^\lambda$, $b' \in \{0, 1\}$ and quantum states $|\psi\rangle_{\mathcal{D} \otimes \mathcal{C}}$ on a λ -qubit register $\mathcal{D}$ and an arbitrary size register $\mathcal{C}$, let the following hold for all QPT distinguishers $\mathcal{D}$:*

$$\left| \Pr \left[\mathcal{D} \left(1^\lambda, \mathcal{Z}(\theta, b', \mathcal{D}), \mathcal{C} \right) = 1 \right] - \Pr \left[\mathcal{D} \left(1^\lambda, \mathcal{Z}(0^\lambda, b', \mathcal{D}), \mathcal{C} \right) = 1 \right] \right| \leq \text{negl}(\lambda)$$

In other words, $\mathcal{Z}$ is semantically-secure wrt θ . Consider now the following experiment wrt an adversary $\mathcal{A}$:

$\text{Exp}_{\mathcal{Z}, \mathcal{A}}(b)$:

1. *The experiment samples $y, \theta \leftarrow \{0, 1\}^\lambda$ and initializes $\mathcal{A}$ with input 1^λ along with the following input:*

$$\mathcal{Z} \left(\theta, b \oplus \bigoplus_{i: \theta_i=0} y_i, |y\rangle_\theta \right)$$

2. $\mathcal{A}$ sends a string $\text{cert} \in \{0, 1\}^\lambda$ and a quantum state ρ_B on register B .
3. If $\forall i \in [\lambda]$ such that $\theta_i = 1$, $\text{cert}_i = y_i$, then the experiment outputs ρ_B . Else, it outputs $\perp$.

Then, the following guarantee holds for all QPT adversaries $\mathcal{A}$:

$$\text{TD}\left(\text{Exp}_{\mathcal{Z}, \mathcal{A}}(0), \text{Exp}_{\mathcal{Z}, \mathcal{A}}(1)\right) = \text{negl}(\lambda)$$

4 Certified-Everlasting NIZK Proofs

4.1 Common Reference String Model

Definition 4.1 (CE-NIZK in the CRS Model). A certified-everlasting non-interactive zero-knowledge proof (CE-NIZK) in the CRS model for a language $\mathcal{L}$ is a tuple of four algorithms ($\text{Setup}, \text{P}, \text{V}, \text{Cert}$) with the following syntax:

- $\text{Setup}(1^\lambda) \rightarrow \text{crs}$: The classical setup algorithm takes as input a security parameter, and outputs a common reference string (CRS) crs .
- $\text{P}(\text{crs}, x, \omega) \rightarrow (\sigma, \rho_P)$: The quantum prover's algorithm takes as input a CRS crs , a statement $x \in \mathcal{L}$, and a witness $\omega \in R_{\mathcal{L}}(x)$. It outputs a quantum proof state σ and a residual state ρ_P .
- $\text{V}(\text{crs}, x, \sigma) \rightarrow (b, \rho_V)$: The quantum verifier's algorithm takes as input the CRS crs , a statement $x \in \mathcal{L}$, and a quantum proof σ . It outputs a bit b along with a residual state ρ_V .
- $\text{Cert}(\rho_P, \rho_V) \rightarrow \top/\perp$: The certification algorithm takes as input a prover's state ρ_P and a verifier's state ρ_V . It outputs $\top$ (accept) or $\perp$ (reject).

A CE-NIZK must satisfy the following properties:

- *Completeness*: The following holds for every $x \in \mathcal{L}$ and $\omega \in R_{\mathcal{L}}(x)$:

$$\Pr \left[\text{V}(\text{crs}, x, \sigma)[1] \rightarrow 0 : \begin{array}{l} \text{crs} \leftarrow \text{Setup}(1^\lambda) \\ (\sigma, \rho_P) \leftarrow \text{P}(\text{crs}, x, \omega) \end{array} \right] \leq \text{negl}(\lambda)$$

- *Deletion Correctness*: The following holds for every $x \in \mathcal{L}$ and $\omega \in R_{\mathcal{L}}(x)$:

$$\Pr \left[\text{Cert}(\rho_P, \rho_V) \rightarrow \perp : \begin{array}{l} \text{crs} \leftarrow \text{Setup}(1^\lambda) \\ (\sigma, \rho_P) \leftarrow \text{P}(\text{crs}, x, \omega) \\ (b, \rho_V) \leftarrow \text{V}(\text{crs}, x, \sigma) \end{array} \right] \leq \text{negl}(\lambda)$$

- *Statistical Soundness*: For every $\tilde{x} \notin \mathcal{L}$, the following holds:

$$\Pr_{\text{crs} \leftarrow \text{Setup}(1^\lambda)} [\exists \tilde{\sigma} : \text{V}(\text{crs}, \tilde{x}, \tilde{\sigma})[1] \rightarrow 1] \leq \text{negl}(\lambda).$$

- *Computational Zero-Knowledge*: There exists a simulator Sim such that for all $x \in \mathcal{L}$ and $\omega \in R_{\mathcal{L}}(x)$, the following holds:

$$\left[(x, \text{crs}, \sigma) \approx_c (x, \text{crs}', \sigma') : \begin{array}{l} \text{crs} \leftarrow \text{Setup}(1^\lambda) \\ (\sigma, \rho_P) \leftarrow \text{P}(\text{crs}, x, \omega) \\ (\text{crs}', \sigma') \leftarrow \text{Sim}(x) \end{array} \right]$$

- *Certified-Everlasting Zero-Knowledge (CE-ZK)*: For every malicious QPT verifier V^* , there exists a QPT simulator Sim_{V^*} such that the following holds for every $x \in \mathcal{L}$ and $\omega \in R_{\mathcal{L}}(x)$:

$$\left[\begin{array}{l} \text{crs} \leftarrow \text{Setup}(1^\lambda) \\ (\sigma, \rho_P) \leftarrow P(\text{crs}, x, \omega) \\ \rho_{A \otimes B} \leftarrow V^*(\text{crs}, x, \sigma) \\ \text{If } \text{Cert}(\rho_P, \rho_A) = \top \text{ then } \tilde{\rho} := \rho_B. \\ \text{Else } \tilde{\rho} := \perp. \\ \rho' \leftarrow \text{Sim}_{V^*}(x) \end{array} \right]$$

Here, $\rho_{A \otimes B}$ is a possibly entangled state on two registers A and B. We denote the residual state on A (likewise B) at any point as ρ_A (likewise ρ_B).

Remark 4.2. *The CE-ZK definition describes non-black-box simulation (for generality in Theorem 2.2), but our feasibility uses only black-box simulation.*

4.2 Shared EPR Model

Definition 4.3 (CE-NIZK in the Shared EPR Model). Let $(P_1, V_1) \dots (P_\ell, V_\ell)$ be registers denoting halves of ℓ -many EPR pairs. Let the prover's register be defined as $P := P_1 \otimes \dots \otimes P_\ell$ and the verifier's register as $V := V_1 \otimes \dots \otimes V_\ell$. A CE-NIZK in the shared EPR model for a language $\mathcal{L} \in \text{NP}$ is a tuple of five algorithms ($\text{Setup}, P, V, \text{Del}, \text{Cert}$) with the following syntax:

- $\text{Setup}(1^\lambda) \rightarrow \text{crs}$: The classical setup algorithm takes as input a security parameter, and outputs a common reference string (CRS) crs .
- $P(\text{crs}, P, x, \omega) \rightarrow (\pi, \rho_P)$: The quantum prover's algorithm takes as input a CRS crs , the prover register P , a statement $x \in \mathcal{L}$, and a witness $\omega \in R_{\mathcal{L}}(x)$. It outputs a classical proof π and a residual state ρ_P on a register $P' \otimes P$ where P is the EPR register and P' is a new register.
- $V(\text{crs}, V, x, \pi) \rightarrow (b, \rho_V)$: The quantum verifier's algorithm takes as input a CRS crs , the verifier register V , a statement $x \in \mathcal{L}$, and a classical proof π . It outputs a bit b along with a residual state ρ_V on register $V' \otimes V$ where V is the EPR register and V' is a new register.
- $\text{Del}(\rho_V) \rightarrow (\text{cert}, \tilde{\rho}_V)$: The quantum deletion algorithm takes as input a state ρ_V on register $V' \otimes V$, and outputs a classical certificate cert , and state $\tilde{\rho}_V$.
- $\text{Cert}(\text{cert}, \rho_P) \rightarrow \top/\perp$: The certification algorithm takes as input a classical certificate cert and a quantum state ρ_P on $P' \otimes P$. It outputs $\top$ or $\perp$.

A CE-NIZK in the shared EPR model must satisfy the following properties:

- *Completeness*: The following holds for every $x \in \mathcal{L}$ and $\omega \in R_{\mathcal{L}}(x)$:

$$\Pr \left[V(\text{crs}, V, x, \pi)[1] \rightarrow 0 : \begin{array}{l} \text{crs} \leftarrow \text{Setup}(1^\lambda) \\ (\pi, \rho_P) \leftarrow P(\text{crs}, P, x, \omega) \end{array} \right] \leq \text{negl}(\lambda)$$

- *Deletion Correctness*: The following holds for every $x \in \mathcal{L}$ and $\omega \in R_{\mathcal{L}}(x)$:

$$\Pr \left[\begin{array}{l} \text{Cert}(\text{cert}, \rho_P) \rightarrow \perp : \\ \text{crs} \leftarrow \text{Setup}(1^\lambda) \\ (\pi, \rho_P) \leftarrow P(\text{crs}, P, x, \omega) \\ (b, \rho_V) \leftarrow V(\text{crs}, V, x, \pi) \\ (\text{cert}, \tilde{\rho}_V) \leftarrow \text{Del}(\rho_V) \end{array} \right] \leq \text{negl}(\lambda)$$

- *Statistical Soundness*: For every $\tilde{x} \notin \mathcal{L}$ and every unbounded P^* , we have:

$$\Pr \left[V(\text{crs}, V, \tilde{x}, \pi^*)[1] \rightarrow 1 : \begin{array}{l} \text{crs} \leftarrow \text{Setup}(1^\lambda) \\ (\pi^*, \rho_P^*) \leftarrow P^*(\text{crs}, P, \tilde{x}) \end{array} \right] \leq \text{negl}(\lambda)$$

- *Computational Zero-Knowledge*: There exists a simulator Sim such that for all $x \in \mathcal{L}$ and $\omega \in R_{\mathcal{L}}(x)$, the following holds:

$$\left[\begin{array}{l} (x, \text{crs}, V, \pi) \approx_c (x, \text{crs}', V', \pi') : \\ \text{crs} \leftarrow \text{Setup}(1^\lambda) \\ (\pi, \rho_P) \leftarrow P(\text{crs}, P, x, \omega) \\ (\text{crs}', V', \pi') \leftarrow \text{Sim}(x) \end{array} \right]$$

- *Certified-Everlasting Zero-Knowledge (CE-ZK)*: There exists a QPT simulator Sim such that for every malicious verifier V^* , statement $x \in \mathcal{L}$ and witness $\omega \in R_{\mathcal{L}}(x)$, the following holds:

$$\left[\begin{array}{l} \tilde{\rho} \approx_s \rho' : \\ \text{crs} \leftarrow \text{Setup}(1^\lambda) \\ (\pi, \rho_P) \leftarrow P(\text{crs}, P, x, \omega) \\ (\text{cert}^*, \rho^*) \leftarrow V^*(\text{crs}, V, x, \pi) \\ \text{If } \text{Cert}(\text{cert}^*, \rho_P) = \top \text{ then } \tilde{\rho} := \rho^*. \\ \text{Else } \tilde{\rho} := \perp. \\ \rho' \leftarrow \text{Sim}^{V^*}(x) \end{array} \right]$$

Here, ρ^* is a state on $V \otimes V'$ where V' is a new register, and Sim^{V^*} denotes black-box access to V^* on its input and setup registers.

5 CE-NIZK in the CRS Model

5.1 Construction from Quantum-Secure NIZK

We now construct a CE-NIZK $\Pi = (\text{Setup}, P, V, \text{Cert})$ as follows:

Certified-Everlasting NIZK in the CRS Model

Building Blocks:

1. NIZK $\Pi_{\text{nizk}} := (\text{Setup}_{\text{nizk}}, P_{\text{nizk}}, V_{\text{nizk}})$ for NP in the CRS model with:
 - Statistical Adaptive Soundness
 - Quantum-Secure Adaptive Zero-Knowledge
 - Non-Adaptive ZK with Statistical CRS Indistinguishability
2. BB84 States $|y\rangle^\theta := H^{\theta_1} |y_1\rangle \otimes \dots \otimes H^{\theta_\lambda} |y_\lambda\rangle$, where $y, \theta \in \{0, 1\}^\lambda$.

3. Quantum-Secure PRF $\{F_{\text{prfk}} : \{0, 1\}^{\text{poly}(\lambda)} \rightarrow \{0, 1\}^{\text{poly}(\lambda)}\}_{\text{prfk} \in \{0, 1\}^\lambda}$.

Setup(1^λ) :

- Compute $\text{crs}_{\text{in}} \leftarrow \text{Setup}_{\text{nizk}}(1^\lambda)$ and $\text{crs}_{\text{out}} \leftarrow \text{Setup}_{\text{nizk}}(1^\lambda)$.
- Output $\text{crs} := \text{crs}_{\text{in}} \parallel \text{crs}_{\text{out}}$.

P(crs, x, ω) :

1. Let ℓ be an upper bound on the proof size of Π_{nizk} . For each $i \in [2\ell]$, sample $y^i, \theta^i \leftarrow \{0, 1\}^\lambda$. Then, construct the following BB84 state on register $R^i := R_1^i \otimes \dots \otimes R_\lambda^i$.

$$|y^i\rangle_{R^i}^{\theta^i} := H^{\theta_1^i} |y_1^i\rangle_{R_1^i} \otimes \dots \otimes H^{\theta_\lambda^i} |y_\lambda^i\rangle_{R_\lambda^i}$$

2. Parse $\text{crs} = \text{crs}_{\text{in}} \parallel \text{crs}_{\text{out}}$. Then, compute $\pi_{\text{in}} \leftarrow P_{\text{nizk}}(\text{crs}_{\text{in}}, x, \omega)$.
3. Sample $k^0, k^1 \leftarrow \{0, 1\}^\ell$ and compute $\text{ct}^0 := \pi_{\text{in}} \oplus \text{pad}^0 \oplus k^0$ and $\text{ct}^1 := 0^\ell \oplus \text{pad}^1 \oplus k^1$, where:

$$\text{pad}^1 := \bigoplus_{i:\theta_i^1=0} y_i^1 \parallel \dots \parallel \bigoplus_{i:\theta_i^\ell=0} y_i^\ell; \quad \text{pad}^0 := \bigoplus_{i:\theta_i^{\ell+1}=0} y_i^{\ell+1} \parallel \dots \parallel \bigoplus_{i:\theta_i^{2\ell}=0} y_i^{2\ell}$$

4. For each $z \in \{0, 1\}^{2\ell\lambda}$, let $z := z^1 \parallel \dots \parallel z^{2\ell}$ where $|z^i| = \lambda$ for each $i \in [2\ell]$. Then, define the following NP statement x^z :

Statement x^z : There exist $\{\theta^i\}_{i \in [2\ell]}$ and $k^0, k^1 \in \{0, 1\}^\ell$ s.t.

$$V_{\text{nizk}}(\text{crs}_{\text{in}}, x, \text{ct}^0 \oplus k^0 \oplus \widetilde{\text{pad}}^0) = 1 \bigvee V_{\text{nizk}}(\text{crs}_{\text{in}}, x, \text{ct}^1 \oplus k^1 \oplus \widetilde{\text{pad}}^1) = 1$$

$$\widetilde{\text{pad}}^1 := \bigoplus_{i:\theta_i^1=0} z_i^1 \parallel \dots \parallel \bigoplus_{i:\theta_i^\ell=0} z_i^\ell; \quad \widetilde{\text{pad}}^0 := \bigoplus_{i:\theta_i^{\ell+1}=0} z_i^{\ell+1} \parallel \dots \parallel \bigoplus_{i:\theta_i^{2\ell}=0} z_i^{2\ell}$$

5. For each $i \in \{0, 1\}^{2\ell\lambda}$, sample $s^{i,0}, s^{i,1} \leftarrow \{0, 1\}^\lambda$. For each $z \in \{0, 1\}^{2\ell\lambda}$:

$$\text{sig}^z := s^{1,z[1]} \parallel \dots \parallel s^{2\ell\lambda, z[2\ell\lambda]}$$

6. Define the register $R := R^1 \otimes \dots \otimes R^{2\ell}$. Let the state on register R be $\sum_z \alpha^z |z\rangle$. Then, compute the following state $|\psi\rangle$ on registers R, P and S :

$$|\psi\rangle_{R \otimes P \otimes S} := \sum_z \alpha^z |z\rangle_R |\pi_{\text{out}}^z\rangle_P |\text{sig}^z\rangle_S$$

where $\pi_{\text{out}}^z \leftarrow P_{\text{nizk}}(\text{crs}_{\text{out}}, x^z, \omega_{\text{out}}; r^z)$, $\omega_{\text{out}} := (\{\theta^i\}_{i \in [2\ell]}, k^0, k^1)$ and explicit randomness $r^z := F_{\text{prfk}}(z)$ for a PRF key sampled as $\text{prfk} \leftarrow \{0, 1\}^\lambda$.

7. Output the quantum proof $\sigma := (|\psi\rangle_{R \otimes P \otimes S}, \text{ct}^0, \text{ct}^1)$ and the verification state $\rho_P := (\omega_{\text{out}}, \text{crs}, \{y^i\}_{i \in [2\ell]}, \text{prfk}, \{s^{i,0}, s^{i,1}\}_{i \in [2\ell\lambda]}, \text{ct}^0, \text{ct}^1)$.

$V(\text{crs}, x, \sigma) :$

1. Parse crs as $\text{crs} = \text{crs}_{\text{in}} \parallel \text{crs}_{\text{out}}$ and $\sigma = (|\psi\rangle_{\text{R} \otimes \text{P} \otimes \text{S}}, \text{ct}^0, \text{ct}^1)$.
2. Let $|\psi\rangle_{\text{R} \otimes \text{P} \otimes \text{S}} = \sum_{\mathbf{z}} \alpha^{\mathbf{z}} |\mathbf{z}\rangle_{\text{R}} |\pi_{\text{out}}^{\mathbf{z}}\rangle_{\text{P}} |\text{sig}^{\mathbf{z}}\rangle_{\text{S}}$. Compute the following state on registers $\text{R}, \text{P}, \text{S}, \text{OUT}$:

$$|\psi\rangle'_{\text{R} \otimes \text{P} \otimes \text{S} \otimes \text{OUT}} := \sum_{\mathbf{z}} \alpha^{\mathbf{z}} |\mathbf{z}\rangle_{\text{R}} |\pi_{\text{out}}^{\mathbf{z}}\rangle_{\text{P}} |\text{sig}^{\mathbf{z}}\rangle_{\text{S}} |V_{\text{nizk}}(\text{crs}_{\text{out}}, x^{\mathbf{z}}, \pi_{\text{out}}^{\mathbf{z}})\rangle_{\text{OUT}}$$

3. Measure the register OUT in the computational basis to obtain outcome $b \in \{0, 1\}$ and a residual state $|\tilde{\psi}\rangle_{\text{R} \otimes \text{P} \otimes \text{S}}$.
4. Set $\rho_{\text{V}} := |\tilde{\psi}\rangle_{\text{R} \otimes \text{P} \otimes \text{S}}$. Finally, output (b, ρ_{V}) .

$\text{Cert}(\rho_{\text{P}}, \rho_{\text{V}}) :$

1. Parse $\rho_{\text{P}} = (\omega_{\text{out}}, \text{crs}, \{y^i\}_{i \in [2\ell]}, \text{prfk}, \{s^{i,0}, s^{i,1}\}_{i \in [2\ell\lambda]}, \text{ct}^0, \text{ct}^1)$ and $\rho_{\text{V}} = |\tilde{\psi}\rangle_{\text{R} \otimes \text{P} \otimes \text{S}}$.
2. Perform $|\mathbf{z}\rangle_{\text{R}} |\widetilde{\text{sig}}^{\mathbf{z}}\rangle_{\text{S}} |0\rangle_{\text{T}} \mapsto |\mathbf{z}\rangle_{\text{R}} |\widetilde{\text{sig}}^{\mathbf{z}}\rangle_{\text{S}} |0 \oplus \text{Test}(\mathbf{z}, \widetilde{\text{sig}}^{\mathbf{z}})\rangle_{\text{T}}$ where T is initialized to 0 and $\text{Test}(\mathbf{z}, \widetilde{\text{sig}}^{\mathbf{z}})$ is a function that outputs 1 if $\text{sig}^{\mathbf{z}} = \widetilde{\text{sig}}^{\mathbf{z}}$ and 0 otherwise.
3. Perform $|\mathbf{z}\rangle_{\text{R}} |\mathbf{t}\rangle_{\text{P}} |\mathbf{s}\rangle_{\text{S}} \mapsto |\mathbf{z}\rangle_{\text{R}} |\mathbf{t} \oplus \text{P}_{\text{nizk}}(\text{crs}_{\text{out}}, x^{\mathbf{z}}, \omega_{\text{out}}; r^{\mathbf{z}})\rangle_{\text{P}} |\mathbf{s} \oplus \text{sig}^{\mathbf{z}}\rangle_{\text{S}}$ where $r^{\mathbf{z}} = \text{F}_{\text{prfk}}(\mathbf{z})$ to get the state $|\tilde{\phi}\rangle_{\text{R} \otimes \text{P} \otimes \text{S}}$.
4. Measure the R register of $|\tilde{\phi}\rangle_{\text{R} \otimes \text{P} \otimes \text{S}}$ in the Hadamard basis to get cert .
5. Let $\theta := \theta^1 \parallel \dots \parallel \theta^{2\ell}$ and $y := y^1 \parallel \dots \parallel y^{2\ell}$. If $\text{cert}_j = y_j$ for each $j \in [2\ell\lambda]$ such that $\theta_j = 1$, then output $\top$. Otherwise, output $\perp$.

Theorem 5.1. *Assuming Π_{nizk} is a NIZK proof for NP with statistical adaptive soundness, quantum-secure (adaptive) zero-knowledge and computational (non-adaptive) zero-knowledge with statistical CRS indistinguishability (Definition 3.6), and $\{\text{F}_{\text{prfk}}\}_{\text{prfk}}$ is a quantum-secure PRF family (Definition 3.2) then $\Pi := (\text{Setup}, \text{P}, \text{V}, \text{Cert})$ is a secure CE-NIZK scheme for NP (Definition 4.1).*

5.2 Security Proof

We now proceed to prove that Π satisfies the following properties of CE-NIZK:

Completeness: Observe that $V_{\text{nizk}}(\text{crs}_{\text{in}}, x, \text{ct}^0 \oplus \widetilde{\text{pad}}^0 \oplus \mathbf{k}^0)$ outputs 1 for each superposition term $\mathbf{z}$. This is because unmasking ct^0 with $\mathbf{k}^0$ will result in $\pi_{\text{in}} \oplus \text{pad}^0$. Moreover, $\widetilde{\text{pad}}^0$ corresponding to every superposition term $\mathbf{z}$ is equal to pad^0 . Next, observe that $V_{\text{nizk}}(\text{crs}_{\text{in}}, x, \pi_{\text{in}})$ outputs 1 with overwhelming probability by the completeness of Π_{nizk} . Consequently, with all but negligible probability, all the statements $x^{\mathbf{z}}$ in superposition are valid. Finally, by completeness of Π_{nizk} , we have that $V_{\text{nizk}}(\text{crs}_{\text{out}}, x^{\mathbf{z}}, \pi_{\text{out}}^{\mathbf{z}})$ outputs 1 with overwhelming probability for each term $\mathbf{z}$, assuming the randomness for computing $\pi_{\text{out}}^{\mathbf{z}}$ is obtained from a quantum-secure PRF F_{prfk} . Consequently,

computing the result of verification on register OUT and measuring the result produces outcome 1 with overwhelming probability.

Deletion-Correctness: Recall that V is sent the state $|\psi\rangle_{R \otimes P \otimes S} = \sum_z \alpha^z |z\rangle_R |\pi_{\text{out}}^z\rangle_P |\text{sig}^z\rangle_S$ and the post-verification state $|\tilde{\psi}\rangle$ is close in trace distance to $|\psi\rangle$ due to the above completeness argument and the gentle measurement lemma. Next, observe that the signature checks using **Test** pass in superposition for every term z and hence the state is undisturbed. Then, **Cert** uncomputes the proofs on registers P, S , resulting in the state $|\tilde{\phi}\rangle$ that is close in trace distance to the state $|y\rangle^\theta$. Consequently, measuring the registers of $|\tilde{\phi}\rangle$ in the Hadamard basis results in a value **cert** that is consistent with y at all the Hadamard basis positions with overwhelming probability. As a result, **Cert** outputs 1 with overwhelming probability.

Computational Zero-Knowledge: This follows immediately from the fact that Π_{nizk} has a simulator Sim_{nizk} such that the values $(\text{crs}_{\text{Sim}}, \pi_{\text{Sim}}) \leftarrow \text{Sim}_{\text{nizk}}(x)$ are computationally indistinguishable from $\text{crs}_{\text{in}}, \pi_{\text{in}}$. Consequently, we can replace π_{in} inside ct^0 with π_{Sim} and crs_{in} (which is part of crs) with crs_{Sim} .

Statistical Soundness: For every false statement x' , with overwhelming probability over the choice of crs_{in} , we know that there doesn't exist any π' such that $V_{\text{nizk}}(\text{crs}_{\text{in}}, x', \pi') = 1$ due to the statistical soundness of Π_{nizk} . Consequently, with overwhelming probability over crs_{in} , all the corresponding statements x_z in superposition are false as well. Now, these statements are decided after crs_{out} is chosen. Consequently, we rely on the adaptive soundness property to ensure that with overwhelming probability over crs_{out} , there do not exist pairs $(x^z, \pi_{\text{out}}^z)$ that would be accepted by $V_{\text{nizk}}(\text{crs}_{\text{out}}, x^z, \pi_{\text{out}}^z)$. Consequently, verification outputs 0 with overwhelming probability, ensuring statistical soundness.

Certified-Everlasting Zero-Knowledge: Consider the following sequence of hybrids:

Hyb₀ :

- V^* is initialized with x and $\text{crs} = \text{crs}_{\text{in}} \parallel \text{crs}_{\text{out}} \leftarrow \text{Setup}(1^\lambda)$.
- Then, V^* is provided with $\sigma = (|\psi\rangle_{R \otimes P \otimes S}, \text{ct}^0, \text{ct}^1)$ computed as $(\sigma, \rho_P) \leftarrow P(\text{crs}, x, \omega)$, where $\text{ct}^0 = \pi_{\text{in}} \oplus \text{pad}^0 \oplus k^0$ and $\text{ct}^1 = 0^\ell \oplus \text{pad}^1 \oplus k^1$.
- Then, V^* outputs a bi-partite state $\rho_{A \otimes B}$. If $\text{Cert}(\rho_P, \rho_A) = \top$, output ρ_B where ρ_A is the state on register A and ρ_B is the residual state on register B. Output $\perp$ otherwise.

Hyb₁ :

- V^* is initialized with x and $\text{crs}_{\text{Sim}} \parallel \text{crs}_{\text{out}}$, where $(\text{crs}_{\text{Sim}}, \pi_{\text{Sim}}) \leftarrow \text{Sim}_{\text{nizk}}(x)$ and $\text{crs}_{\text{out}} \leftarrow \text{Setup}_{\text{nizk}}(1^\lambda)$, with Sim_{nizk} being the QPT simulator of Π_{nizk} .
- Then, V^* is provided with $\sigma = (|\psi\rangle_{R \otimes P \otimes S}, \text{ct}^0, \text{ct}^1)$ computed as $(\sigma, \rho_P) \leftarrow P(\text{crs}_{\text{Sim}} \parallel \text{crs}_{\text{out}}, x, \omega)$, where $\text{ct}^0 = \pi_{\text{in}} \oplus \text{pad}^0 \oplus k^0$ and $\text{ct}^1 = 0^\ell \oplus \text{pad}^1 \oplus k^1$.
- Then, V^* outputs a bi-partite state $\rho_{A \otimes B}$. If $\text{Cert}(\rho_P, \rho_A) = \top$, output ρ_B where ρ_A is the state on register A and ρ_B is the residual state on register B. Output $\perp$ otherwise.

$\text{Hyb}_0 \approx_s \text{Hyb}_1$ follows from the statistical CRS indistinguishability property corresponding to Sim_{nizk} , which ensures that $(x, \text{crs}_{\text{in}}) \approx_s (x, \text{crs}_{\text{Sim}})$.

Consider now the hybrids, $\text{Hyb}_2^1, \dots, \text{Hyb}_2^\ell$ where Hyb_2^i corresponds to changing the i -th bit of 0^ℓ inside ct^1 into the i -th bit of π_{Sim} . Note that this also changes the statements x^z for the superposition terms z corresponding to the R register of $|\psi\rangle_{R \otimes P \otimes S}$. Consequently, the hybrids also involve a change to the proofs π_{out}^z in superposition on register P accordingly. Now, let $\text{Hyb}_2^0 := \text{Hyb}_1$ and observe that Hyb_2^ℓ is the following hybrid:

Hyb_2^ℓ :

- V^* is initialized with x and $\text{crs}_{\text{Sim}} \parallel \text{crs}_{\text{out}}$, where $(\text{crs}_{\text{Sim}}, \pi_{\text{Sim}}) \leftarrow \text{Sim}_{\text{nizk}}(x)$ and $\text{crs}_{\text{out}} \leftarrow \text{Setup}_{\text{nizk}}(1^\lambda)$, with Sim_{nizk} being the QPT simulator of Π_{nizk} .
- Then, V^* is provided with $\sigma = (|\psi\rangle_{R \otimes P \otimes S}, \text{ct}^0, \text{ct}^1)$ computed as $(\sigma, \rho_P) \leftarrow P(\text{crs}_{\text{Sim}} \parallel \text{crs}_{\text{out}}, x, \omega)$, except that $\text{ct}^0 = \pi_{\text{in}} \oplus \text{pad}^0 \oplus k^0$ and $\text{ct}^1 = \pi_{\text{Sim}} \oplus \text{pad}^1 \oplus k^1$.
- Then, V^* outputs a bi-partite state $\rho_{A \otimes B}$. If $\text{Cert}(\rho_P, \rho_A) = \top$, output ρ_B where ρ_A is the state on register A and ρ_B is the residual state on register B . Output $\perp$ otherwise.

Lemma 5.2. $\forall i \in [\ell] : \text{Hyb}_2^{i-1} \approx_s \text{Hyb}_2^i$

Proof. Let us now focus on the i -th BB84 state $|y^i\rangle_{R^i}^{\theta^i}$, and the distribution $\mathcal{Z}^i$:

$\mathcal{Z}^i(\theta^i, b', D)$:

- If $\theta^i \neq 0^\lambda$, then execute the following:
 1. Compute ct^0, ct^1 in the same way as in Hyb_2^{i-1} except b' is set as the i -th bit inside ct^1 . In particular, we have:
$$\text{ct}^0 := \pi_{\text{in}} \oplus \text{pad}^0 \oplus k^0, \quad \text{ct}^1 := \pi_{\text{Sim}}[1] \parallel \dots \parallel \pi_{\text{Sim}}[i-1] \parallel b' \parallel 0^{\ell-i} \oplus \text{pad}^1 \oplus k^1$$
 2. Compute $|\psi\rangle_{R \otimes P \otimes S}$ in the same way as in Hyb_2^{i-1} , except using θ^i from the input (instead of sampling it) and register D instead of sampling the state $|y^i\rangle^{\theta^i}$.
 3. Initialize V^* with input $x, \text{crs}_{\text{Sim}} \parallel \text{crs}_{\text{out}}$ and provide it with the proof $\sigma := (|\psi\rangle_{R \otimes P \otimes S}, \text{ct}^0, \text{ct}^1)$.
 4. When V^* outputs a bi-partite state $\rho_{A \otimes B}$, parse it as a state $\rho_{R \otimes P \otimes S \otimes B}$ on registers R, P, S and B .
 5. Perform the map $|z\rangle_R |\widetilde{\text{sig}}^z\rangle_S |0\rangle_T \mapsto |z\rangle_R |\widetilde{\text{sig}}^z\rangle_S |0 \oplus \text{Test}(z, \widetilde{\text{sig}}^z)\rangle_T$ where Test is as defined in the construction. Then, measure register T and output $\perp$ if the result is not 1.
 6. Perform $|z\rangle_R |t\rangle_P |s\rangle_S \mapsto |z\rangle_R |t \oplus P_{\text{nizk}}(\text{crs}_{\text{out}}, x^z, \omega_{\text{out}}; r^z)\rangle_P |s \oplus \text{sig}^z\rangle_S$ on registers R, P and S where $r^z = F_{\text{prfk}}(z)$ to get the state $\rho'_{R \otimes P \otimes S \otimes B}$.
 7. Output the state $\rho'_{R \otimes P \otimes S \otimes B}$ along with $\{\theta^j, y^j\}_{j \in [2\ell] \setminus \{i\}}$.
- If $\theta^i = 0^\lambda$, then execute the following:
 1. Compute the state $\sum_z \alpha^z |z\rangle_R$ in the same way as in Hyb_2^{i-1} , except that register D is used instead of the state $|y^i\rangle^{\theta^i}$.

2. Compute ct^0, ct^1 as random strings, i.e.:

$$\text{ct}^0 \leftarrow \{0, 1\}^\ell, \quad \text{ct}^1 \leftarrow \{0, 1\}^\ell$$

3. Then, compute the state $|\psi\rangle_{\text{R} \otimes \text{P} \otimes \text{S}}$ as:

$$|\psi\rangle_{\text{R} \otimes \text{P} \otimes \text{S}} := \sum_{\mathbf{z}} \alpha^{\mathbf{z}} |\mathbf{z}\rangle_{\text{R}} |\pi_{\mathbf{S}}^{\mathbf{z}}\rangle_{\text{P}} |\text{sig}^{\mathbf{z}}\rangle_{\text{S}}$$

where the NIZK proof $\pi_{\mathbf{S}}^{\mathbf{z}}$ is computed as $\pi_{\mathbf{S}}^{\mathbf{z}} \leftarrow S_2(\text{td}, x^{\mathbf{z}}; r^{\mathbf{z}})$ where $r^{\mathbf{z}} := F_{\text{prfk}}(\mathbf{z})$ for $\text{prfk} \leftarrow \{0, 1\}^\lambda$ and $(\text{crs}_{\mathbf{S}}, \text{td}) \leftarrow S_1(1^\lambda)$ where $\mathbf{S} := (S_1, S_2)$ is the quantum-secure adaptive zero-knowledge simulator of Π_{nizk} .

4. Initialize V^* with input $x, \text{crs}_{\text{Sim}} \parallel \text{crs}_{\mathbf{S}}$ and provide it with the proof $\sigma := (|\psi\rangle_{\text{R} \otimes \text{P} \otimes \text{S}}, \text{ct}^0, \text{ct}^1)$.
5. When V^* outputs a bi-partite state $\rho_{\text{A} \otimes \text{B}}$, parse it as a state $\rho_{\text{R} \otimes \text{P} \otimes \text{S} \otimes \text{B}}$.
6. Perform the map $|\mathbf{z}\rangle_{\text{R}} |\widetilde{\text{sig}}^{\mathbf{z}}\rangle_{\text{S}} |0\rangle_{\text{T}} \mapsto |\mathbf{z}\rangle_{\text{R}} |\widetilde{\text{sig}}^{\mathbf{z}}\rangle_{\text{S}} |0 \oplus \text{Test}(\mathbf{z}, \widetilde{\text{sig}}^{\mathbf{z}})\rangle_{\text{T}}$ where Test is as defined in the construction. Then, measure register T and output $\perp$ if the result is not 1.
7. Perform the map $|\mathbf{z}\rangle_{\text{R}} |\mathbf{t}\rangle_{\text{P}} |\mathbf{s}\rangle_{\text{S}} \mapsto |\mathbf{z}\rangle_{\text{R}} |\mathbf{t} \oplus S_2(\text{td}, x^{\mathbf{z}}; r^{\mathbf{z}})\rangle_{\text{P}} |\mathbf{s} \oplus \text{sig}^{\mathbf{z}}\rangle_{\text{S}}$ on registers R, P and S where $r^{\mathbf{z}} = F_{\text{prfk}}(\mathbf{z})$ to get the state $\rho'_{\text{R} \otimes \text{P} \otimes \text{S} \otimes \text{B}}$.
8. Output the state $\rho'_{\text{R} \otimes \text{P} \otimes \text{S} \otimes \text{B}}$ along with $\{\theta^j, y^j\}_{j \in [2\ell] \setminus \{i\}}$.

Lemma 5.3. For all $\theta^i \in \{0, 1\}^\lambda, b' \in \{0, 1\}$ and states $|\tau\rangle_{\text{C} \otimes \text{D}}$ on registers C, D where D is of size λ , the following holds:

$$(\text{C}, \mathcal{Z}^i(\theta^i, b', \text{D})) \approx_c (\text{C}, \mathcal{Z}^i(0^\lambda, b', \text{D}))$$

Proof. Consider the following hybrids that act on inputs (θ^i, b', D) :

- $\underline{\mathcal{Z}}_0^i$: Same as $\mathcal{Z}^i$.
- $\underline{\mathcal{Z}}_1^i$: Similar to $\mathcal{Z}_0^i$, except that after the signature check in Step 5., another check is performed before Step 6., where the terms $\mathbf{z}$ in superposition are post-selected to be valid statements wrt the witness $\omega_{\text{out}} = (\{\theta^i\}_{i \in [2\ell]}, k^0, k^1)$. In other words, the map $|\mathbf{z}\rangle_{\text{R}} |0\rangle_{\text{W}} \mapsto |\mathbf{z}\rangle_{\text{R}} |\text{Check}(\mathbf{z})\rangle_{\text{W}}$ is first performed, where $\text{Check}(\mathbf{z})$ is a function that outputs 1 iff $\omega_{\text{out}} \in R_{\text{out}}(x^{\mathbf{z}})$, where R_{out} is the NP relation specified by the description of $x^{\mathbf{z}}$. Then, the W register is measured and $\perp$ is output if the result is not 1.
- $\underline{\mathcal{Z}}_2^i$: Similar to $\mathcal{Z}_1^i$, except that it generates simulated proofs $\pi_{\mathbf{S}}^{\mathbf{z}}$ on register P as in the distribution $\mathcal{Z}(0^\lambda, b', \text{D})$, instead of the real proofs $\pi_{\text{out}}^{\mathbf{z}}$, apart from replacing crs_{out} with a simulated reference string $\text{crs}_{\mathbf{S}}$.
- $\underline{\mathcal{Z}}_3^i$: Similar to $\mathcal{Z}_2^i$, except that ct^0, ct^1 are sampled as truly random values, and the simulated proofs $\pi_{\mathbf{S}}^{\mathbf{z}}$ are generated according to the new ciphertexts.
- $\underline{\mathcal{Z}}_4^i$: Similar to $\mathcal{Z}_3^i$, except that the post-selection step introduced in $\mathcal{Z}_1^i$ is no longer performed.

We begin by arguing the indistinguishability of $\mathcal{Z}_0^i$ and $\mathcal{Z}_1^i$ as follows:

Lemma 5.4. For all $\theta^i \in \{0, 1\}^\lambda$, $b' \in \{0, 1\}$ and states $|\tau\rangle_{\mathcal{C} \otimes \mathcal{D}}$ with a λ -size register $\mathcal{D}$, we have:

$$(\mathcal{C}, \mathcal{Z}_0^i(\theta^i, b', \mathcal{D})) \approx_s (\mathcal{C}, \mathcal{Z}_1^i(\theta^i, b', \mathcal{D}))$$

Proof. The intuition here is that if the hybrids are not close, then one can use the one-way-to-hiding (O2H) lemma [AHU19] in order to efficiently find a value $\tilde{z}$ along with a valid signature/MAC $\widetilde{\text{sig}}^z$ such that ω_{out} is not a valid witness wrt the statement $x^{\tilde{z}}$. Since the statements z in superposition given to $\mathcal{V}^*$ are all valid wrt ω_{out} , the last $\ell\lambda$ bits of $\tilde{z}$ must differ from $y^{\ell+1} \parallel \dots \parallel y^{2\ell}$ at least one computational basis position. It can then be shown that an adversary cannot guess the corresponding “block” of the signature $\widetilde{\text{sig}}^z$, except with negligible probability. We show this formally as follows:

Consider first the case when $\theta^i \neq 0^\lambda$ and assume that there exists an unbounded distinguisher $\mathcal{D}$ that has non-negligible advantage in distinguishing between the distributions. Consider the following functions G, H :

$G[\text{aux}](z, \widetilde{\text{sig}}^z)$:

- Output $(1, \{s^{i,0}, s^{i,1}\}_{i \in [2\ell\lambda]})$ if $\text{Test}(z, \widetilde{\text{sig}}^z) = 1$ AND $\omega_{\text{out}} \in \mathcal{R}_{\text{out}}(x^z)$, where $\text{Test}, x^z, \omega_{\text{out}}, \{s^{i,0}, s^{i,1}\}_{i \in [2\ell\lambda]}$ and $\mathcal{R}_{\text{out}}$ are all as defined in Section 5, and are parsed from aux that is part of the description of G .
- Otherwise, output $(0, \{s^{i,0}, s^{i,1}\}_{i \in [2\ell\lambda]})$.

$H[\text{aux}](z, \widetilde{\text{sig}}^z)$:

- Output $(1, \{s^{i,0}, s^{i,1}\}_{i \in [2\ell\lambda]})$ if $\text{Test}(z, \widetilde{\text{sig}}^z) = 1$ where $\text{Test}, x^z, \omega_{\text{out}}, \{s^{i,0}, s^{i,1}\}_{i \in [2\ell\lambda]}$ and $\mathcal{R}_{\text{out}}$ are all as defined in Section 5, and are parsed from aux that is part of the description of H .
- Otherwise, output $(0, \{s^{i,0}, s^{i,1}\}_{i \in [2\ell\lambda]})$.

We now construct an algorithm $\mathcal{A}$ that makes use of $\mathcal{D}$ in order to distinguish between the oracles G and H .

$\mathcal{A}^\mathcal{O}(\rho_{\text{aux}})$:

- From the input ρ_{aux} , obtain the state $\rho_{\mathcal{R} \otimes \mathcal{P} \otimes \mathcal{S} \otimes \mathcal{B}}$, which corresponds to the state computed after Step 4. of $\mathcal{Z}^i$.
- Evaluate $\mathcal{O}$ (which is either G or H) in superposition of registers $\mathcal{R}, \mathcal{S}$ and post-select on the outcome being 1. Also, obtain $\{s^{i,0}, s^{i,1}\}_{i \in [2\ell\lambda]}$ from $\mathcal{O}$ and all other values corresponding to $\mathcal{Z}^i$ from the auxiliary input state ρ_{aux} .
- Perform the uncomputation step, i.e., Step 6. of $\mathcal{Z}^i$ to get the state $\rho'_{\mathcal{R} \otimes \mathcal{P} \otimes \mathcal{S} \otimes \mathcal{B}}$.
- Finally, run $\mathcal{D}$ on input $\rho'_{\mathcal{R} \otimes \mathcal{P} \otimes \mathcal{S} \otimes \mathcal{B}}$ and $\{\theta^j, y^j\}_{j \in [2\ell] \setminus \{i\}}$ and output the bit output by $\mathcal{D}$.

Observe that $\mathcal{A}^\mathcal{O}$ provides $\mathcal{D}$ with either the output of $\mathcal{Z}_0^i$ or $\mathcal{Z}_1^i$ depending on whether $\mathcal{O}$ is G or H .

Consider now the following algorithm $\mathcal{R}$ which is given a state ρ_{aux} which consists of the state $\rho_{\mathcal{R} \otimes \mathcal{P} \otimes \mathcal{S} \otimes \mathcal{B}}$ that is computed after Step 4. of $\mathcal{Z}^i$ along with other values sampled in $\mathcal{Z}^i$. Importantly, $\mathcal{R}$ is only given $s^{j,y[j]}$ for each $j \in [(\ell+1)\lambda, 2\ell\lambda]$ such that $\theta_j = 0$. Then, $\mathcal{R}$ runs the algorithm $\mathcal{A}^H(\rho_{\text{aux}})$, except it doesn't respond to the query and simply measures the oracle query made by $\mathcal{A}$ to obtain $(\tilde{z}, \widetilde{\text{sig}}^z)$. Observe that $\mathcal{R}$ here simply corresponds to the algorithm $\mathcal{B}^H$ guaranteed by the

O2H Lemma in the special case when $q = 1$. Hence, with non-negligible probability, $\tilde{z}$ is such that $\omega_{\text{out}} \notin R_{\text{out}}(x^{\tilde{z}})$ and yet $\widetilde{\text{sig}}^{\tilde{z}}$ is a valid signature, i.e., $\widetilde{\text{sig}}^{\tilde{z}} = \text{sig}^{\tilde{z}}$. Clearly, this means there exists some $j \in [(\ell + 1)\lambda, 2\ell\lambda]$ where $\theta_j = 0$ and $\tilde{z}[j] \neq y[j]$. Notice however, that $\mathcal{R}$ was given no information about $s^{j, 1-y[j]}$. Consequently, it is easy to see by a union bound analysis that $\widetilde{\text{sig}}^{\tilde{z}} = \text{sig}^{\tilde{z}}$ holds with at most negligible probability, a contradiction. This means no such distinguisher D exists, and the proof of the $\theta = 0^\lambda$ case is essentially the same, finishing the proof. $\square$

Next, we argue that $\mathcal{Z}_1^i$ and $\mathcal{Z}_2^i$ are computationally close. Recall that by definition, the statements x^z corresponding to superposition terms z are all true wrt ω_{out} before computation of the proofs on register P . Moreover, the same holds before uncomputation as well, due to the post-selection step introduced in $\mathcal{Z}_1^i$. Hence, we can invoke quantum-secure adaptive zero-knowledge (Definition 3.6) of Π_{nizk} to guarantee that $(C, \mathcal{Z}_1^i(\theta^i, b', D)) \approx_c (C, \mathcal{Z}_2^i(\theta^i, b', D))$, as the two only differ in that one uses real proofs, while the other uses simulated proofs. In particular, both the computation and uncomputation of the proofs in the former (likewise latter) distribution can be performed using quantum oracle access (as in Definition 3.6) to $P(\text{crs}_{\text{out}}, \cdot, \cdot)$ (likewise $S_2(\text{td}, \cdot)$). Note that by quantum-security of F_{prfk} , we can assume wlog that the oracle access to $P(\text{crs}_{\text{out}}, \cdot, \cdot), S_2(\text{td}, \cdot)$ in the quantum-secure adaptive zero-knowledge definition to be simulated via randomness derived from the PRF.

Observe now that $(C, \mathcal{Z}_2^i(\theta^i, b', D)) \equiv (C, \mathcal{Z}_3^i(\theta^i, b', D))$ because in the former, the ciphertexts ct^0, ct^1 are masked by uniform and independent values k^0, k^1 , while they are directly sampled from the uniform distribution in the latter. The equivalence of the distributions then follows from the fact that the simulated proofs are directly computed using the statements x^z (which depend on ct^0, ct^1).

Finally, $(C, \mathcal{Z}_3^i(\theta^i, b', D)) \approx_s (C, \mathcal{Z}_4^i(\theta^i, b', D))$ follows from essentially the same argument as that of Lemma 5.4. Notice now that $\mathcal{Z}_4^i(\theta^i, b', D)$ is the same distribution as $\mathcal{Z}^i(0^\lambda, b', D)$, and recall that $\mathcal{Z}_0^i(\theta^i, b', D)$ is the same distribution as $\mathcal{Z}^i(\theta^i, b', D)$. Therefore, we have $(C, \mathcal{Z}^i(\theta^i, b', D)) \approx_c (C, \mathcal{Z}^i(0^\lambda, b', D))$. $\square$

Next, assume for contradiction that there exists a malicious QPT verifier V^* and an unbounded distinguisher D that distinguishes the output of Hyb_2^{i-1} and Hyb_2^i with non-negligible probability. We will now construct a QPT adversary $\mathcal{A}$ (along with an unbounded distinguisher) that participates in experiment $\text{Exp}_{\mathcal{Z}^i, \mathcal{A}}$ specified by Theorem 3.9 and breaks its certified-deletion security.

$\text{Exp}_{\mathcal{Z}^i, \mathcal{A}}(b)$:

1. Sample $y^i, \theta^i \leftarrow \{0, 1\}^\lambda$ and initialize $\mathcal{A}$ with 1^λ along with the following input (or the input $\perp$ if $\mathcal{Z}^i$ does output $\perp$):

$$(\rho'_{R \otimes P \otimes S \otimes B}, \{\theta^j, y^j\}_{j \in [2\ell] \setminus \{i\}}) := \mathcal{Z}^i\left(\theta^i, b \oplus \bigoplus_{j: \theta_j^i = 0} y_j^i, |y^i\rangle^{\theta^i}\right)$$

2. $\mathcal{A}$ executes as follows:

- Measure the R register of $\rho'_{R \otimes P \otimes S \otimes B}$ in the Hadamard basis to get outcome cert and leftover state $\rho''_{R \otimes P \otimes S \otimes B}$.
- Parse $\text{cert} = \text{cert}^1 \parallel \dots \parallel \text{cert}^{2\ell}$ where $\text{cert}^j \in \{0, 1\}^\lambda$ for each $j \in [2\ell]$.
- If for each $j \in [2\ell] \setminus \{i\}$, it holds that $\text{cert}_k^j = y_k^j$ for each $k \in [\lambda]$ such that $\theta_k^j = 1$, then output cert^i along with the state ρ''_B on register B . Otherwise, set $\rho''_B := \perp$ and output cert^i along with ρ''_B .

3. If $\forall j \in [\lambda]$ such that $\theta_j^i = 1$, $\text{cert}_j^i = y_j^i$, output ρ_B'' (and $\perp$ otherwise).

Observe now that the output of Hyb_2^{i-1} is identically distributed to the output of $\text{Exp}_{\mathcal{Z}^i, \mathcal{A}}(0)$ (conditioned on $\theta^i \neq 0^\lambda$ which occurs with $\text{negl}(\lambda)$ probability). Likewise, Hyb_2^i is statistically close to $\text{Exp}_{\mathcal{Z}^i, \mathcal{A}}(1)$, assuming a bit flip occurs from Hyb_2^{i-1} to Hyb_2^i (otherwise, the hybrids are identically distributed). Hence, the unbounded distinguisher D can also distinguish between $\text{Exp}_{\mathcal{Z}^i, \mathcal{A}}(0)$ and $\text{Exp}_{\mathcal{Z}^i, \mathcal{A}}(1)$ with non-negligible advantage. However, Theorem 3.9 guarantees that $\text{Exp}_{\mathcal{Z}^i, \mathcal{A}}(0) \approx_s \text{Exp}_{\mathcal{Z}^i, \mathcal{A}}(1)$, a contradiction. Therefore, $\text{Hyb}_2^{i-1} \approx_s \text{Hyb}_2^i$. $\square$

As a consequence of the above lemma, we have that $\text{Hyb}_1 \approx_s \text{Hyb}_2^\ell$.

Next, consider the hybrids $\text{Hyb}_2^{\ell+1}, \dots, \text{Hyb}_2^{2\ell}$, in which the proof π_{in} inside ct^0 will be replaced bit-by-bit with 0s. In other words, $\text{Hyb}_2^{2\ell}$ is the following:

$\text{Hyb}_2^{2\ell}$:

- V^* is initialized with $\text{crs}_{\text{Sim}} \parallel \text{crs}_{\text{out}}$, where $(\text{crs}_{\text{Sim}}, \pi_{\text{Sim}}) \leftarrow \text{Sim}_{\text{nizk}}(x)$ and $\text{crs}_{\text{out}} \leftarrow \text{Setup}_{\text{nizk}}(1^\lambda)$, with Sim_{nizk} being the QPT simulator of Π_{nizk} .
- Then, V^* is provided with $\sigma = (|\psi\rangle_{\mathbb{R} \otimes \mathbb{P} \otimes \mathbb{S}}, \text{ct}^0, \text{ct}^1)$ computed as $(\sigma, \rho_P) \leftarrow P(\text{crs}_{\text{Sim}} \parallel \text{crs}_{\text{out}}, x, \omega)$, except that $\text{ct}^0 = \text{red}^\ell \oplus \text{pad}^0 \oplus \text{k}^0$ and $\text{ct}^1 = \pi_{\text{Sim}} \oplus \text{pad}^1 \oplus \text{k}^1$.
- Then, V^* outputs a bi-partite state $\rho_{A \otimes B}$. If $\text{Cert}(\rho_P, \rho_A) = \top$, output ρ_B where ρ_A is the state on register A and ρ_B is the residual state on register B . Output $\perp$ otherwise.

Lemma 5.5. $\forall i \in [\ell] : \text{Hyb}_2^{\ell+i-1} \approx_s \text{Hyb}_2^{\ell+i}$

Proof. This follows from a similar argument as the proof of Lemma 5.2. This is due to the fact that the statement x^z remains true whether a particular bit inside ct^0 is flipped or not. This is because ct^1 already consists of π_{Sim} which is sufficient to satisfy the second clause of the OR condition of statement x^z , due to the computational zero-knowledge property of Π_{nizk} . Consequently, we can invoke the quantum-secure adaptive zero-knowledge guarantee for the outer proofs as in the proof of Lemma 5.2. $\square$

Using this lemma, we have $\text{Hyb}_2^\ell \approx_s \text{Hyb}_2^{2\ell}$, which gives us that $\text{Hyb}_0 \approx_s \text{Hyb}_2^{2\ell}$. Notice that in $\text{Hyb}_2^{2\ell}$, the witness ω is never used and that the view of V^* can be simulated entirely with the statement x . This finishes the proof of CE-ZK, and hence the proof of Π being a CE-NIZK in the CRS model.

6 CE-NIZK in the Shared EPR Model

6.1 The Hidden-Bits Paradigm

Definition 6.1 (Hidden Bits Generator [QRW19]). A hidden bits generator (HBG) is a triple of algorithms $(\text{Setup}, \text{GenBits}, \text{Ver})$ with the following syntax:

- $\text{Setup}(1^\lambda, 1^k) \rightarrow \text{crs}$: The setup algorithm takes as input a security parameter and the number of hidden-bits k , and outputs a common reference string crs .
- $\text{GenBits}(\text{crs}) \rightarrow (\text{com}, r, \{\pi_i\}_{i \in [k]})$: The hidden bits generation algorithm outputs a commitment com , hidden bits $r \in \{0, 1\}^k$ and openings $\{\pi_i\}_{i \in [k]}$.
- $\text{Ver}(\text{crs}, \text{com}, i, r_i, \pi_i) \rightarrow \top / \perp$: The verification algorithm takes an index i , hidden bit r_i and proof π_i along with a CRS crs and commitment com . It outputs $\top$ (accept) or $\perp$ (reject).

An HBG must satisfy the following properties:

- *Correctness*: For all $i \in [k]$, the following holds:

$$\Pr \left[\text{Ver}(\text{crs}, \text{com}, i, r_i, \pi_i) = \perp : \begin{array}{l} \text{crs} \leftarrow \text{Setup}(1^\lambda, 1^k) \\ (\text{com}, r, \pi_{[k]}) \leftarrow \text{GenBits}(\text{crs}) \end{array} \right] \leq \text{negl}(\lambda)$$

- *Succinct Commitment*: There exists a set of commitments $\mathcal{COM}(\lambda)$ such that for all $\text{crs} \leftarrow \text{Setup}(1^\lambda, 1^k)$ and every com output by $\text{GenBits}(\text{crs})$, $\text{com} \in \mathcal{COM}(\lambda)$. Moreover, there exists a constant $\delta < 1$ such that $|\mathcal{COM}(\lambda)| \leq 2^{k^\delta \text{poly}(\lambda)}$. Furthermore, $\forall \text{com} \notin \mathcal{COM}(\lambda)$, $\text{Ver}(\text{crs}, \text{com}, \cdot, \cdot, \cdot)$ outputs $\perp$.
- *Statistical Binding*: There exists a (possibly inefficient) deterministic algorithm $\text{Open}(1^k, \text{crs}, \text{com})$ such that for all $k = \text{poly}(\lambda)$, crs and com , it outputs r such that for every every unbounded $\tilde{P}$, the following holds:

$$\Pr \left[\begin{array}{l} r_i^* \neq r_i \\ \wedge \\ \text{Ver}(\text{crs}, \text{com}, i, r_i^*, \pi_i) = \top \end{array} : \begin{array}{l} \text{crs} \leftarrow \text{Setup}(1^\lambda, 1^k) \\ (\text{com}, i, r_i^*, \pi_i) \leftarrow \tilde{P}(\text{crs}) \\ r \leftarrow \text{Open}(1^k, \text{crs}, \text{com}) \end{array} \right] \leq \text{negl}(\lambda)$$

- *Computational Hiding*: For all $k = \text{poly}(\lambda)$ and $I \subseteq [k]$, the following holds:

$$\left[\begin{array}{l} (\text{crs}, \text{com}, I, r_I, \pi_I, r_{\bar{I}}) \\ \approx_c \\ (\text{crs}, \text{com}, I, r_I, \pi_I, r'_{\bar{I}}) \end{array} : \begin{array}{l} \text{crs} \leftarrow \text{Setup}(1^\lambda, 1^k) \\ (\text{com}, r, \pi_{[k]}) \leftarrow \text{GenBits}(\text{crs}) \\ r' \leftarrow \{0, 1\}^k \end{array} \right]$$

Theorem 6.2 ([WWW25]). *Assuming the polynomial hardness of LWE, there exists an HBG with statistical binding and computational hiding in the CRS model.*

Definition 6.3 (NIZK in the Hidden Bits Model [FLS90]). A NIZK proof in the hidden bits model (HBM) for a language $\mathcal{L} \in \text{NP}$ is a pair of algorithms (P, V) with the following syntax, where $k(\lambda, n) = \text{poly}(\lambda, n)$:

- $P(r, x, \omega) \rightarrow (I, \pi)$: The prover takes a hidden bit string $r \in \{0, 1\}^{k(\lambda, n)}$, a statement x with size $|x| = n$, and a witness ω as input. It outputs a set of indices $I \subseteq [k]$ and a proof π .
- $V(I, r_I, x, \pi) \rightarrow \top / \perp$: The verifier on input a set of indices I , hidden bits $r_I = \{r_i\}_{i \in I}$, statement x and proof π , outputs $\top / \perp$.

Such a NIZK must satisfy the following properties:

- *Completeness*: For every $x \in \mathcal{L}$ of size $|x| = n$ and $\omega \in R_{\mathcal{L}}(x)$, we have:

$$\Pr \left[V(I, r_I, x, \pi) \rightarrow 0 : \begin{array}{l} r \leftarrow \{0, 1\}^{k(\lambda, n)} \\ (I, \pi) \leftarrow P(r, x, \omega) \end{array} \right] \leq \text{negl}(\lambda)$$

- *Soundness*: For all $n = \text{poly}(\lambda)$ and every unbounded P^* , the following holds:

$$\Pr \left[V(I, r_I, x, \pi) \rightarrow 1 \wedge |x| = n \wedge x \notin \mathcal{L} : \begin{array}{l} r \leftarrow \{0, 1\}^{k(\lambda, n)} \\ (x, \pi, I) \leftarrow P^*(r) \end{array} \right] \leq \text{negl}(\lambda)$$

- *Zero-Knowledge*: There exists a simulator Sim such that for every $x \in \mathcal{L}$ and $\omega \in R_{\mathcal{L}}(x)$, the following holds:

$$\left[\begin{array}{lcl} (x, I, r_I, \pi) \approx_s (x, I', r'_I, \pi') & : & \begin{array}{l} r \leftarrow \{0, 1\}^{k(\lambda, n)} \\ (I, \pi) \leftarrow \text{P}(r, x, \omega) \\ (I', r'_I, \pi') \leftarrow \text{Sim}(x) \end{array} \end{array} \right]$$

Theorem 6.4 ([FLS90]). *NIZKs exist in the HBM for every language $\mathcal{L} \in \text{NP}$.*

Remark 6.5 (Amplifying Soundness [QRW19]). *For any polynomial $q(\lambda, n)$, given a NIZK in the HBM (requiring k' many hidden bits) with $\text{negl}(\lambda)$ soundness error, we can build one with $2^{-q(\lambda, n)}$ $\text{negl}(\lambda)$ soundness error. This is obtained by a q -fold parallel repetition of the base NIZK, requiring $q \cdot k'$ many hidden bits.*

6.2 Construction from any Hidden Bits Generator

Certified-Everlasting NIZK in the Shared EPR Model

Building Blocks:

1. Hidden Bits Generator $\text{HBG} := (\text{Setup}_{\text{bg}}, \text{GenBits}_{\text{bg}}, \text{Ver}_{\text{bg}})$.
2. NIZK in the Hidden Bits Model $\Pi_{\text{hb}} := (\text{P}_{\text{hb}}, \text{V}_{\text{hb}})$.
3. Let ℓ, k be polynomials in λ , where ℓ is assumed to be sufficiently large (the exact size is determined in the proof of statistical soundness, based on the number of hidden bits k' required by Π_{hb}). For each $(i, j) \in [\ell] \times [k]$, consider EPR pairs of the form $\text{P}_j^i, \text{V}_j^i$. For each $i \in [\ell]$, set $\text{P}^i := \text{P}_1^i \otimes \dots \otimes \text{P}_k^i$ and likewise for V^i . Finally, set $\text{P} := \text{P}^1 \otimes \dots \otimes \text{P}^\ell$ and likewise for V .

Setup(1^λ):

- Sample $\text{crs}_{\text{bg}} \leftarrow \text{Setup}_{\text{bg}}(1^\lambda, 1^{k\ell})$ and $s \leftarrow \{0, 1\}^\ell$.
- Output $\text{crs} := (\text{crs}_{\text{bg}}, s)$.

P(crs, P, x, ω):

1. Parse $\text{crs} = (\text{crs}_{\text{bg}}, s)$ and $\text{P} = \text{P}^1 \otimes \dots \otimes \text{P}^\ell$.
2. Compute $(\text{com}, \theta, \{\text{op}^{(i,j)}\}_{(i,j) \in [\ell] \times [k]}) \leftarrow \text{GenBits}_{\text{bg}}(\text{crs}_{\text{bg}})$, where $\theta = \{\theta^i\}_{i \in [\ell]}$ and $\theta^i \in \{0, 1\}^k$ for each $i \in [\ell]$, while $\text{op}^{(i,j)}$ denotes the opening for bit θ_j^i .
3. For each $i \in [\ell]$, execute the following:
 - (a) Measure P^i in the θ^i basis to get outcome y^i .
 - (b) Compute $t_i := \bigoplus_{j: \theta^i[j]=0} y_j^i$.
4. Compute $r_i := t_i \oplus s_i$ for each $i \in [\ell]$.

5. Execute $(I \subseteq [\ell], \pi_{\text{hb}}) \leftarrow \text{P}_{\text{hb}}(r, x, \omega)$.

6. Output $\pi := (I, \pi_{\text{hb}}, \text{com}, \theta^I = \{\theta^i\}_{i \in I}, \{\text{op}^{(i,j)}\}_{(i,j) \in I \times [k]})$ and $\rho_{\text{P}} := (y^1 \parallel \dots \parallel y^\ell, \theta, I, \text{P})$.

$\text{V}(\text{crs}, \text{V}, x, \pi) :$

1. Parse $\pi = (I, \pi_{\text{hb}}, \text{com}, \theta^I = \{\theta^i\}_{i \in I}, \{\text{op}^{(i,j)}\}_{i \in I, j \in [k]})$, $\text{crs} = (\text{crs}_{\text{bg}}, s)$ and $\text{V} = \text{V}^1 \otimes \dots \otimes \text{V}^\ell$.

2. For each $i \in I$, do the following:

- For each $j \in [k]$, if $\text{Ver}_{\text{bg}}(\text{crs}_{\text{bg}}, \text{com}, (i, j), \theta_j^i, \text{op}^{(i,j)}) \neq \top$, output $\perp$.
- Measure V^i in the θ^i basis to get y^i .
- Compute $t_i := \bigoplus_{j: \theta_j^i = 0} y_j^i$.

3. Compute $r_I := t_I \oplus s_I$.

4. Output the bit $b := \text{V}_{\text{hb}}(I, r_I, x, \pi_{\text{hb}})$ along with $\rho_{\text{V}} := (I, \text{V})$.

$\text{Del}(\rho_{\text{V}}) :$

1. Parse $\rho_{\text{V}} = (I, \text{V})$ and $\text{V} = \text{V}^1 \otimes \dots \otimes \text{V}^\ell$.

2. For each $i \in [\ell] \setminus I$, measure V^i in the Hadamard basis to get cert^i .

3. Output $\text{cert} := \{\text{cert}^i\}_{i \in [\ell] \setminus I}$.

$\text{Cert}(\text{cert}, \rho_{\text{P}}) :$

1. Parse $\text{cert} = \{\text{cert}^i\}_{i \in [\ell] \setminus I}$, $\rho_{\text{P}} = (y^1 \parallel \dots \parallel y^\ell, \theta, I, \text{P})$.

2. For each $i \in [\ell] \setminus I$ and $j \in [k]$ such that $\theta_j^i = 1$, if $\text{cert}_j^i = y_j^i$, output $\top$. Otherwise, output $\perp$.

Theorem 6.6. *Let HBG be any statistical-binding hidden bits generator and Π_{hb} be a NIZK for NP in the hidden bits model. Then, $\Pi = (\text{Setup}, \text{P}, \text{V}, \text{Del}, \text{Cert})$ is a CE-NIZK for NP in the shared EPR model.*

Hence, from Theorem 6.2 and Theorem 6.4, we have the following:

Corollary 6.7. *Assuming the polynomial hardness of LWE, there exists a CE-NIZK for NP in the shared EPR model.*

Proof. We show that Π satisfies the properties of CE-NIZK as follows:

Completeness: The hidden-bits NIZK proof π_{hb} is computed as $(I, \pi_{\text{hb}}) \leftarrow \text{P}_{\text{hb}}(r, x, \omega)$ where $r = t \oplus s$. Clearly, the distribution of r is uniform. By correctness of the HBG HBG, all of the $\text{Ver}_{\text{bg}}(\text{crs}_{\text{bg}}, \text{com}, (i, j), \theta_j^i, \text{op}^{(i,j)})$ checks will pass. Moreover, the verifier V computes the correct r_I values as s_I is available as part of crs and the t_I values are common due to the EPR pairs being

measured in the same basis on both ends for the indices corresponding to I . Finally, $V_{\text{hb}}(I, r_I, x, \pi_{\text{hb}})$ outputs 1 by the completeness of the hidden-bits NIZK Π_{hb} . Consequently, V outputs 1.

Deletion Correctness: This follows directly from the fact that for each $i \in [\ell] \setminus I$ and each $j \in [k]$ such that $\theta_j^i = 1$, both parties measure in the Hadamard basis to get $\text{cert}_j^i = y_j^i$.

Computational Zero-Knowledge: For every $x \in \mathcal{L}$ and $\omega \in R_{\mathcal{L}}(x)$, consider the following hybrids:

Hyb₀^{zk} : This is the real experiment, which proceeds as follows:

1. $\text{crs}_{\text{bg}} \leftarrow \text{Setup}_{\text{bg}}(1^\lambda, 1^{k\ell})$. $s \leftarrow \{0, 1\}^\ell$.
2. $(\text{com}, \theta, \{\text{op}^{(i,j)}\}_{(i,j) \in [\ell] \times [k]}) \leftarrow \text{GenBits}_{\text{bg}}(\text{crs}_{\text{bg}})$.
3. For each $i \in [\ell]$, compute t_i as in Π from register P using θ as the measurement basis. Let $t = t_1 \parallel \dots \parallel t_\ell$.
4. Compute $r := t \oplus s$.
5. Execute $(I, \pi_{\text{hb}}) \leftarrow P_{\text{hb}}(r, x, \omega)$.
6. Output $\text{crs} := (\text{crs}_{\text{bg}}, s)$, $\pi := (I, \pi_{\text{hb}}, \text{com}, \{\theta^i\}_{i \in I}, \{\text{op}^{(i,j)}\}_{(i,j) \in I \times [k]})$ and the verifier register V .

Hyb₁^{zk} : Here, r is sampled before s as follows:

1. $\text{crs}_{\text{bg}} \leftarrow \text{Setup}_{\text{bg}}(1^\lambda, 1^{k\ell})$. $r \leftarrow \{0, 1\}^\ell$.
2. $(\text{com}, \theta, \{\text{op}^{(i,j)}\}_{(i,j) \in [\ell] \times [k]}) \leftarrow \text{GenBits}_{\text{bg}}(\text{crs}_{\text{bg}})$.
3. For each $i \in [\ell]$, compute t_i as in Π from register P using θ as the measurement basis. Let $t = t_1 \parallel \dots \parallel t_\ell$.
4. Compute $s := t \oplus r$.
5. Execute $(I, \pi_{\text{hb}}) \leftarrow P_{\text{hb}}(r, x, \omega)$.
6. Output $\text{crs} := (\text{crs}_{\text{bg}}, s)$, $\pi := (I, \pi_{\text{hb}}, \text{com}, \{\theta^i\}_{i \in I}, \{\text{op}^{(i,j)}\}_{(i,j) \in I \times [k]})$ and the verifier register V .

Clearly, $\text{Hyb}_0^{\text{zk}} \equiv \text{Hyb}_1^{\text{zk}}$ as the outputs are identically distributed.

Hyb₂^{zk} : Here, the EPRs for the unopened indices $i \in [\ell] \setminus I$ are measured in a random basis.

1. $\text{crs}_{\text{bg}} \leftarrow \text{Setup}_{\text{bg}}(1^\lambda, 1^{k\ell})$. $r \leftarrow \{0, 1\}^\ell$.
2. $(\text{com}, \theta, \{\text{op}^{(i,j)}\}_{(i,j) \in [\ell] \times [k]}) \leftarrow \text{GenBits}_{\text{bg}}(\text{crs}_{\text{bg}})$.
3. For each $i \in [\ell] \setminus I$, sample $\tilde{\theta}^i \leftarrow \{0, 1\}^k$.
4. For each $i \in [\ell]$, compute t_i as in Π from register P , using θ^i if $i \in I$ and using $\tilde{\theta}^i$ if $i \notin I$. Let $t = t_1 \parallel \dots \parallel t_\ell$.

5. Compute $s := t \oplus r$.
6. Execute $(I, \pi_{\text{hb}}) \leftarrow \text{P}_{\text{hb}}(r, x, \omega)$.
7. Output $\text{crs} := (\text{crs}_{\text{bg}}, s)$, $\pi := (I, \pi_{\text{hb}}, \text{com}, \{\theta^i\}_{i \in I}, \{\text{op}^{(i,j)}\}_{(i,j) \in I \times [k]})$ and the verifier register V .

Observe that $\text{Hyb}_1^{\text{zk}} \approx_c \text{Hyb}_2^{\text{zk}}$ due to the fact that the values $\{\theta^i\}_{i \in [\ell] \setminus I}$ are pseudo-random, by the computational hiding property of HBG.

Hyb_3^{zk} : Here, s is sampled uniformly for the unopened indices $i \in [\ell] \setminus I$.

1. $\text{crs}_{\text{bg}} \leftarrow \text{Setup}_{\text{bg}}(1^\lambda, 1^{k\ell})$. $r \leftarrow \{0, 1\}^\ell$.
2. $(\text{com}, \theta, \{\text{op}^{(i,j)}\}_{(i,j) \in [\ell] \times [k]}) \leftarrow \text{GenBits}_{\text{bg}}(\text{crs}_{\text{bg}})$.
3. For each $i \in [\ell] \setminus I$, compute $\tilde{\theta}^i \leftarrow \{0, 1\}^k$.
4. For each $i \in [\ell]$, compute t_i as in Π from register P , using θ^i if $i \in I$ and **using $\tilde{\theta}^i$ if $i \notin I$** . Let $t = t_1 \parallel \dots \parallel t_\ell$.
5. Compute $s_I := t_I \oplus r_I$. For $i \in [\ell] \setminus I$, sample **$s_i \leftarrow \{0, 1\}$** .
6. Execute $(I, \pi_{\text{hb}}) \leftarrow \text{P}_{\text{hb}}(r, x, \omega)$.
7. Output $\text{crs} := (\text{crs}_{\text{bg}}, s)$, $\pi := (I, \pi_{\text{hb}}, \text{com}, \{\theta^i\}_{i \in I}, \{\text{op}^{(i,j)}\}_{(i,j) \in I \times [k]})$ and the verifier register V .

Notice that the values $\{\tilde{\theta}^i\}_{i \in [\ell] \setminus I}$ are uniform and independent of the transcript. Hence, the values t_i for $i \in [\ell] \setminus I$ are uniform and independent of the output, by the no-signaling principle. Consequently, we have that $\text{Hyb}_2^{\text{zk}} \equiv \text{Hyb}_3^{\text{zk}}$.

Hyb_4^{zk} : Finally, the values I, r_I, π_{hb} are generated using the hidden bits simulator $\text{Sim}_{\text{hb}}(x)$.

1. $\text{crs}_{\text{bg}} \leftarrow \text{Setup}_{\text{bg}}(1^\lambda, 1^{k\ell})$. **$(I, r_I, \pi_{\text{hb}}) \leftarrow \text{Sim}_{\text{hb}}(x)$** .
2. $(\text{com}, \theta, \{\text{op}^{(i,j)}\}_{(i,j) \in [\ell] \times [k]}) \leftarrow \text{GenBits}_{\text{bg}}(\text{crs}_{\text{bg}})$.
3. For each $i \in [\ell] \setminus I$, compute $\tilde{\theta}^i \leftarrow \{0, 1\}^k$.
4. For each $i \in [\ell]$, compute t_i as in Π from register P , using θ^i if $i \in I$ and **using $\tilde{\theta}^i$ if $i \notin I$** . Let $t = t_1 \parallel \dots \parallel t_\ell$.
5. Compute $s_I := t_I \oplus r_I$. For $i \in [\ell] \setminus I$, sample $s_i \leftarrow \{0, 1\}$.
6. Output $\text{crs} := (\text{crs}_{\text{bg}}, s)$, $\pi := (I, \pi_{\text{hb}}, \text{com}, \{\theta^i\}_{i \in I}, \{\text{op}^{(i,j)}\}_{(i,j) \in I \times [k]})$ and the verifier register V .

Observe that $\text{Hyb}_3^{\text{zk}} \approx_s \text{Hyb}_4^{\text{zk}}$ by the statistical zero-knowledge property of the hidden-bits NIZK Π_{hb} . This gives us that $\text{Hyb}_0^{\text{zk}} \approx_c \text{Hyb}_4^{\text{zk}}$. Furthermore, Hyb_4^{zk} can be simulated entirely with the statement x , without knowledge of the witness ω . Hence, computational zero-knowledge follows.

Certified-Everlasting Zero-Knowledge: For every $x \in \mathcal{L}$ and $\omega \in R_{\mathcal{L}}(x)$, consider the following hybrids:

$\text{Hyb}_0^{\text{cezk}}$: This is the real experiment, which proceeds as follows:

1. $\text{crs}_{\text{bg}} \leftarrow \text{Setup}_{\text{bg}}(1^\lambda, 1^{k\ell})$. $s \leftarrow \{0, 1\}^\ell$.
2. $(\text{com}, \theta, \{\text{op}^{(i,j)}\}_{(i,j) \in [\ell] \times [k]}) \leftarrow \text{GenBits}_{\text{bg}}(\text{crs}_{\text{bg}})$.
3. For each $i \in [\ell]$, compute t_i as in Π from register P using θ as the measurement basis. Let $t = t_1 \| \dots \| t_\ell$.
4. Compute $r := t \oplus s$.
5. Execute $(I, \pi_{\text{hb}}) \leftarrow P_{\text{hb}}(r, x, \omega)$.
6. Compute $\text{crs} := (\text{crs}_{\text{bg}}, s)$, $\pi := (I, \pi_{\text{hb}}, \text{com}, \theta^I = \{\theta^i\}_{i \in I}, \{\text{op}^{(i,j)}\}_{(i,j) \in I \times [k]})$.
7. Execute $(\text{cert}^*, \rho^*) \leftarrow V^*(\text{crs}, V, x, \pi)$.
8. If $\text{Cert}(\text{cert}^*, \rho_P) = \top$, where $\rho_P := (y^1 \| \dots \| y^\ell, \theta, I, P)$ then set $\rho := \rho^*$. Else, set $\rho := \perp$.
9. Output ρ .

$\text{Hyb}_1^{\text{cezk}}$: This is similar to $\text{Hyb}_0^{\text{cezk}}$, except that the bits $\{r_i\}_{i \in [\ell] \setminus I}$ are sampled uniformly at random.

1. $\text{crs}_{\text{bg}} \leftarrow \text{Setup}_{\text{bg}}(1^\lambda, 1^{k\ell})$. $s \leftarrow \{0, 1\}^\ell$.
2. $(\text{com}, \theta, \{\text{op}^{(i,j)}\}_{(i,j) \in [\ell] \times [k]}) \leftarrow \text{GenBits}_{\text{bg}}(\text{crs}_{\text{bg}})$.
3. For each $i \in [\ell]$, compute t_i as in Π from register P using θ as the measurement basis. Let $t = t_1 \| \dots \| t_\ell$.
4. Compute $r_i := t_i \oplus s_i$ for each $i \in I$, and **sample $r_i \leftarrow \{0, 1\}$ for each $i \in [\ell] \setminus I$** .
5. Execute $(I, \pi_{\text{hb}}) \leftarrow P_{\text{hb}}(r, x, \omega)$.
6. Compute $\text{crs} := (\text{crs}_{\text{bg}}, s)$, $\pi := (I, \pi_{\text{hb}}, \text{com}, \theta^I = \{\theta^i\}_{i \in I}, \{\text{op}^{(i,j)}\}_{(i,j) \in I \times [k]})$.
7. Execute $(\text{cert}^*, \rho^*) \leftarrow V^*(\text{crs}, V, x, \pi)$.
8. If $\text{Cert}(\text{cert}^*, \rho_P) = \top$, where $\rho_P := (y^1 \| \dots \| y^\ell, \theta, I, P)$ then set $\rho := \rho^*$. Else, set $\rho := \perp$.
9. Output ρ .

We will now argue that $\text{Hyb}_1^{\text{cezk}} \approx_s \text{Hyb}_0^{\text{cezk}}$ by the certified-deletion property of BB84 states (Section 3.5) due to [BK23]. However, we find that the structure of Theorem 3.9 is incompatible with our setting. Hence, we look into the proof of the main theorem of [BK23] and prove the above claim as follows.

For each $j \in [\ell - |I|]$, consider the hybrid $\text{Hyb}_0^{\text{cezk}, j}$ where the first j bits of r that correspond to indices in $[\ell] \setminus I$ are replaced with truly random bits. Let $\text{Hyb}_0^{\text{cezk}, 0} := \text{Hyb}_0^{\text{cezk}}$ and observe that $\text{Hyb}_0^{\text{cezk}, \ell - |I|} \equiv \text{Hyb}_1^{\text{cezk}}$.

Consider now some arbitrary $j^* \in [\ell - |I|]$. We argue that $\text{Hyb}_0^{\text{cezk}, j^* - 1} \approx_s \text{Hyb}_0^{\text{cezk}, j^*}$, using the following series of experiments:

$\text{Exp}_1(b)$:

- Compute (crs, V, x, π) in the same way as in $\text{Hyb}_0^{\text{cezk}, j^*-1}$, except using b instead of $r[i^*]$.
- If $t[i^*] = b \oplus s[i^*]$, then proceed as in $\text{Hyb}_0^{\text{cezk}, j^*-1}$, i.e., by running V^* and determining the output ρ . Otherwise, output $\perp$.

$\text{Exp}_2(b)$:

- Compute s, r as in $\text{Hyb}_0^{\text{cezk}, j^*-1}$, except that b is used in place of $r[i^*]$. Then, use these values to compute (crs, x, π) and prepare the EPR pairs P, V .
- Run $V^*(\text{crs}, V, x, \pi)$ to get (cert^*, ρ^*) .
- Measure the register P^{i^*} as in $\text{Hyb}_0^{\text{cezk}, j^*-1}$ to compute $t[i^*]$.
- If $t[i^*] = b \oplus s[i^*]$, then proceed as in $\text{Hyb}_0^{\text{cezk}, j^*-1}$, i.e., by running V^* and determining the output ρ . Otherwise, output $\perp$.

Observe that if the trace distance between $\text{Hyb}_0^{\text{cezk}, j^*-1}$ and $\text{Hyb}_0^{\text{cezk}, j^*}$ is non-negligible, then $\text{Adv}(\text{Exp}_1) = \text{TD}(\text{Exp}_1(0), \text{Exp}_1(1)) = \text{non-negl}(\lambda)$. Moreover, notice that $\text{Adv}(\text{Exp}_2) = \text{Adv}(\text{Exp}_1)$, as the measurements on different registers commute.

Next, consider the following experiment:

$\text{Exp}'_2(b)$: This is the same as $\text{Exp}_2(b)$, except that $t[i^*]$ is determined by measuring P^{i^*} in a basis $\tilde{\theta}^{i^*}$ sampled as $\tilde{\theta}^{i^*} \leftarrow \{0, 1\}^k$, instead of the basis θ^{i^*} specified by the HBG commitment com.

It is now easy to see from Claim 3.3 of [BK23] (specifically, the ePrint version [BK22]) that $\text{Adv}(\text{Exp}'_2) = \text{negl}(\lambda)$. Moreover, it can be seen from Claim 3.4 of [BK23] (specifically, the ePrint version [BK22]) that $\text{Adv}(\text{Exp}_2) = \text{negl}(\lambda)$, as otherwise, there exists an efficient reduction that distinguishes between $\tilde{\theta}^{i^*}$ and θ^{i^*} , thereby breaking the computational hiding property of the hidden bits generator HBG. Consequently, we arrive at a contradiction, proving that $\text{Hyb}_0^{\text{cezk}, j^*-1} \approx_s \text{Hyb}_0^{\text{cezk}, j^*}$. Then, we have from a hybrid argument that $\text{Hyb}_0^{\text{cezk}} \approx_s \text{Hyb}_1^{\text{cezk}}$.

$\text{Hyb}_2^{\text{cezk}}$: This is similar to $\text{Hyb}_1^{\text{cezk}}$, except that r is sampled before s .

1. $\text{crs}_{\text{bg}} \leftarrow \text{Setup}_{\text{bg}}(1^\lambda, 1^{k\ell})$. $r \leftarrow \{0, 1\}^\ell$.
2. $(\text{com}, \theta, \{\text{op}^{(i,j)}\}_{(i,j) \in [\ell] \times [k]}) \leftarrow \text{GenBits}_{\text{bg}}(\text{crs}_{\text{bg}})$.
3. For each $i \in [\ell]$, compute t_i as in Π from register P using θ as the measurement basis. Let $t = t_1 \| \dots \| t_\ell$.
4. Compute $s_i := t_i \oplus r_i$ for each $i \in I$, and sample $s_i \leftarrow \{0, 1\}$ for each $i \in [\ell] \setminus I$.
5. Execute $(I, \pi_{\text{hb}}) \leftarrow P_{\text{hb}}(r, x, \omega)$.
6. Compute $\text{crs} := (\text{crs}_{\text{bg}}, s)$, $\pi := (I, \pi_{\text{hb}}, \text{com}, \theta^I = \{\theta^i\}_{i \in I}, \{\text{op}^{(i,j)}\}_{(i,j) \in I \times [k]})$.
7. Execute $(\text{cert}^*, \rho^*) \leftarrow V^*(\text{crs}, V, x, \pi)$.
8. If $\text{Cert}(\text{cert}^*, \rho_P) = \top$, where $\rho_P := (y^1 \| \dots \| y^\ell, \theta, I, P)$ then set $\rho := \rho^*$. Else, set $\rho := \perp$.
9. Output ρ .

Clearly, we have $\text{Hyb}_1^{\text{cezk}} \equiv \text{Hyb}_2^{\text{cezk}}$, as the values are all identically distributed.

$\text{Hyb}_3^{\text{cezk}}$: Finally, the values I, r_I, π_{hb} are generated using the hidden bits simulator $\text{Sim}_{\text{hb}}(x)$.

1. $\text{crs}_{\text{bg}} \leftarrow \text{Setup}_{\text{bg}}(1^\lambda, 1^{k\ell})$. $(I, r_I, \pi_{\text{hb}}) \leftarrow \text{Sim}_{\text{hb}}(x)$.
2. $(\text{com}, \theta, \{\text{op}^{(i,j)}\}_{(i,j) \in [\ell] \times [k]}) \leftarrow \text{GenBits}_{\text{bg}}(\text{crs}_{\text{bg}})$.
3. For each $i \in [\ell]$, compute t_i as in Π from register P using θ as the measurement basis. Let $t = t_1 \parallel \dots \parallel t_\ell$.
4. Compute $s_i := t_i \oplus r_i$ for each $i \in I$, and sample $s_i \leftarrow \{0, 1\}$ for each $i \in [\ell] \setminus I$.
5. Compute $\text{crs} := (\text{crs}_{\text{bg}}, s)$, $\pi := (I, \pi_{\text{hb}}, \text{com}, \theta^I = \{\theta^i\}_{i \in I}, \{\text{op}^{(i,j)}\}_{(i,j) \in I \times [k]})$.
6. Execute $(\text{cert}^*, \rho^*) \leftarrow \text{V}^*(\text{crs}, V, x, \pi)$.
7. If $\text{Cert}(\text{cert}^*, \rho_P) = \top$, where $\rho_P := (y^1 \parallel \dots \parallel y^\ell, \theta, I, P)$ then set $\rho := \rho^*$. Else, set $\rho := \perp$.
8. Output ρ .

Observe that $\text{Hyb}_3^{\text{cezk}} \approx_s \text{Hyb}_2^{\text{cezk}}$ holds because of the statistical zero-knowledge property of Π_{hb} . Moreover, the output of $\text{Hyb}_3^{\text{cezk}}$ can be generated using only x and without the witness ω . Hence, certified-everlasting zero-knowledge holds.

Statistical Soundness: Let P^* be an unbounded malicious prover that breaks soundness with some $\mu(\lambda)$ probability:

$$\Pr \left[\begin{array}{l} \text{V}(\text{crs}, V, x, \pi^*) = \top \\ \wedge \\ x \notin \mathcal{L} \end{array} : \begin{array}{l} \text{crs} \leftarrow \text{Setup}(1^\lambda) \\ (x, \pi^*) \leftarrow P^*(\text{crs}, P) \end{array} \right] \geq \mu(\lambda)$$

where $\pi^* = (I, \pi_{\text{hb}}, \text{com}^*, \{\theta^i\}_{i \in I}, \{\text{op}^{(i,j)}\}_{(i,j) \in I \times [k]})$.

Consider now a verifier $\tilde{V}_0$ that behaves similarly to V , except that it measures all the registers $V^1 \dots V^\ell$ in the computational basis. It is easy to see that if P^* convinces V with probability $\mu(\lambda)$, then it also convinces $\tilde{V}_0$ with probability $\mu(\lambda)$. This is because only the computational basis positions are used for verification by V .

Next, consider a verifier $\tilde{V}$ that behaves similar to $\tilde{V}_0$ except that the registers $V^1, \dots, V^\ell$ are measured in the computational basis before P^* is initialized. Since operations on different registers commute, P^* also convinces $\tilde{V}$ with probability $\mu(\lambda)$:

$$\Pr \left[\begin{array}{l} \tilde{V}(\text{crs}, V, x, \pi^*) = \top \\ \wedge \\ x \notin \mathcal{L} \end{array} : \begin{array}{l} \text{crs} \leftarrow \text{Setup}(1^\lambda) \\ (x, \pi^*) \leftarrow P^*(\text{crs}, P) \end{array} \right] \geq \mu(\lambda)$$

where $\pi^* = (I, \pi_{\text{hb}}, \text{com}^*, \{\theta^i\}_{i \in I}, \{\text{op}^{(i,j)}\}_{(i,j) \in I \times [k]})$.

Now, let $y^1, \dots, y^\ell$ be the values measured by $\tilde{V}$ corresponding to registers $V^1, \dots, V^\ell$. Consequently, once the basis choices $\theta^1, \dots, \theta^\ell$ are specified, the values $t_1, \dots, t_\ell$ are fixed as they are computed as the parity of the computational basis values. Observe now that by statistical binding of HBG, except with negligible probability, we have that r_I computed by $\tilde{V}$ satisfies $r_I = t_I \oplus s_I$ where $t_i = \bigoplus_{j: \tilde{\theta}_j^i = 0} y_j^i$ for each $i \in I$ and $\tilde{\theta}^1 \parallel \dots \parallel \tilde{\theta}^\ell := \text{Open}(1^{k\ell}, \text{crs}_{\text{bg}}, \text{com}^*)$. Moreover, succinctness

of HBG guarantees that $\text{com}^* \in \mathcal{COM}(\lambda)$. Hence, we have the following, where $\text{op}^{I,[k]}$ denotes $\{\text{op}^{(i,j)}\}_{(i,j) \in I \times [k]}$:

$$v(\lambda) := \Pr \left[\begin{array}{l} \mathbf{V}_{\text{hb}}(I, r_I, x, \pi_{\text{hb}}) = \top \\ \wedge x \notin \mathcal{L} \\ \wedge \text{com}^* \in \mathcal{COM}(\lambda) \end{array} : \begin{array}{l} \text{crs} = (\text{crs}_{\text{bg}}, s) \leftarrow \text{Setup}(1^\lambda) \\ (x, \pi^*) \leftarrow \mathbf{P}^*(\text{crs}, \mathbf{P}) \\ \tilde{\theta}^1 \parallel \dots \parallel \tilde{\theta}^\ell := \text{Open}(1^{k\ell}, \text{crs}_{\text{bg}}, \text{com}^*) \\ \forall i \in [\ell] : t_i := \bigoplus_{j: \tilde{\theta}_j^i = 0} y_j^i \\ r = t \oplus s \end{array} \right] \\ \geq \mu(\lambda) - \text{negl}(\lambda)$$

where $\pi^* = (I, \pi_{\text{hb}}, \text{com}^*, \{\theta^i\}_{i \in I}, \{\text{op}^{(i,j)}\}_{(i,j) \in I \times [k]})$.

Next, we define a similar probability for any fixed $\text{com} \in \mathcal{COM}(\lambda)$:

$$v_{\text{com}}(\lambda) := \Pr \left[\begin{array}{l} \mathbf{V}_{\text{hb}}(I, r_I, x, \pi_{\text{hb}}) = \top \\ \wedge x \notin \mathcal{L} \\ \wedge \text{com}^* = \text{com} \end{array} : \begin{array}{l} \text{crs} = (\text{crs}_{\text{bg}}, s) \leftarrow \text{Setup}(1^\lambda) \\ (x, \pi^*) \leftarrow \mathbf{P}^*(\text{crs}, \mathbf{P}) \\ \{\tilde{\theta}^i\}_{i \in [\ell]} := \text{Open}(1^{k\ell}, \text{crs}_{\text{bg}}, \text{com}^*) \\ \forall i \in [\ell] : t_i := \bigoplus_{j: \tilde{\theta}_j^i = 0} y_j^i \\ r = t \oplus s \end{array} \right] \\ \leq \Pr \left[\begin{array}{l} \mathbf{V}_{\text{hb}}(I, r_I, x, \pi_{\text{hb}}) = \top \\ \wedge x \notin \mathcal{L} \end{array} : \begin{array}{l} \text{crs} = (\text{crs}_{\text{bg}}, s) \leftarrow \text{Setup}(1^\lambda) \\ \tilde{\theta}^1 \parallel \dots \parallel \tilde{\theta}^\ell := \text{Open}(1^{k\ell}, \text{crs}_{\text{bg}}, \text{com}) \\ \forall i \in [\ell] : t_i := \bigoplus_{j: \tilde{\theta}_j^i = 0} y_j^i \\ r = t \oplus s \\ (x, I, \pi_{\text{hb}}) \leftarrow \hat{\mathbf{P}}_{\text{crs}_{\text{bg}}, \text{com}}(r) \end{array} \right] \\ \leq 2^{-q(\lambda, n)} \cdot \text{negl}(\lambda)$$

where $\pi^* = (I, \pi_{\text{hb}}, \text{com}^*, \{\theta^i\}_{i \in I}, \{\text{op}^{(i,j)}\}_{(i,j) \in I \times [k]})$.

The first inequality follows because r is already fixed by com and the second follows from the soundness of the hidden bits NIZK by viewing $\hat{\mathbf{P}}_{\text{crs}_{\text{bg}}, \text{com}}(r)$ as a malicious hidden bits model prover and from the fact that r is uniformly random (because s is uniform).

Next, we set $q(\lambda, n)^{1-\delta} = k \cdot k' \cdot p(\lambda)$ so that:

$$(k\ell)^\delta \cdot p(\lambda) = (k \cdot k' \cdot q(\lambda, n))^\delta \cdot p(\lambda) \leq q^\delta(\lambda, n) \cdot q^{1-\delta}(\lambda, n)$$

This ensures that $|\mathcal{COM}(\lambda)| \cdot v_{\text{com}}(\lambda) \leq \text{negl}(\lambda)$. Therefore, a union bound gives us the following:

$$\mu(\lambda) - \text{negl}(\lambda) \leq v(\lambda) \leq \sum_{\text{com} \in \mathcal{COM}(\lambda)} v_{\text{com}}(\lambda) \leq \text{negl}(\lambda)$$

This proves that $\mu(\lambda) \leq \text{negl}(\lambda)$, ensuring soundness. $\square$

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